

**STRAM-PESO: AN ONLINE EMPLOYMENT AND APPLICANT MANAGEMENT SYSTEM
FOR PUBLIC EMPLOYMENT SERVICE OFFICE (LMD-PESO)**

A Project Proposal
Presented to the Faculty of the
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Web Systems and Technologies 2

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STRAM-PESO: An Online Employment and Applicant Management System for Public Employment Service Office (LMD-PESO)

Keywords: *Keywords: Online Employment System, Applicant Management System, Public Employment Service Office (PESO), MERN Stack, REST API, Role-Based Access Control, E-Recruitment, Job Matching System, Real-Time Messaging, Web-Based Application, Centralized Database, Cloud Deployment, JWT Authentication, Resume Management, Applicant Tracking System*

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CHAPTER 1: TECHNICAL DESCRIPTION & PROPOSAL

1.1. Background of the Study

Employment plays an important role in societal well-being and economic stability. It not only provides individuals with secured financial, purpose in life, and opportunities to grow personally and professionally. This idea is closely aligned with the United Nations Sustainable Development Goal 8 (*Decent Work and Economic Growth*) [1], which highlight the importance of promoting sustained, inclusive, and sustainable economic growth, full and productive employment [2].

In the digital era, online employment has emerged as a life-changing platform that expands access to decent work beyond different boundaries. Globally, policies continue to prioritize job creation and workforce development [3], while fast technological advancements change the labor market and work models [4]. Online job seeking has become increasingly well known, with the e-recruitment platforms that address issues like job mismatch and proper qualification, ensuring a more efficient and effective recruitment process [2].

Connectivity to Digital has also fostered a new business model. Technological development and rapid global connectivity are spurring new business models, including planetary labor markets borderless employment opportunities that encourage workers from across the globe to participate in shared employment opportunities. This concept highlights how online employment transcends geographic boundaries, creating a borderless workforce that fosters innovation, reduces unemployment, and contributes to equitable economic growth [5].

The COVID-19 pandemic further accelerated this shift, toward digital economies and remote working models. Where online employment platforms equip people with flexible job options, promote workforce diversification, and respond to evolving skill demands required in modern industries. These also support marginalized and underserved populations by broadening access to job information and employment opportunities beyond their immediate locales, contributing to enhanced economic inclusion and social mobility [6].

This global transformation is reflected in local initiatives such as the Livelihood Manpower Development Public Employment Service Office (LMD-PESO) in Marinduque. Located within the Provincial Capitol Compound in Boac, LMD-PESO provides central employment services. It also provides

timely and efficient services to grassroots workers. Managed by local government units and supported by NGOs and educational institutions, offering job application assistance, recruitment support, and referrals to partner agencies [7].

By addressing issues like job mismatch and proper qualification, LMD-PESO not only helps individuals have livelihoods but also strengthens regional employment stability. Its community-based approach demonstrates how localized online job services can contribute to sustainable economic development, aligning with both national priorities and global goals under SDG 8 [1].

To respond to the challenges identified in the current employment landscape, the STRAM-PESO (Streamlined Recruitment and Applicant Management for Public Employment Service Office) system is proposed. Designed to enhance the efficiency and inclusivity of PESO operations, STRAM-PESO aims to provide a dynamic, accessible, and technology-driven employment platform. It will feature (1) a centralized applicant database to reduce job mismatch and streamline recruitment, (2) automated job posting and matching tools to improve transparency and efficiency, (3) a progress tracking dashboard for both applicants and employers, (4) secure digital transactions to ensure reliability and trust, and (5) integrated reporting functions to support data-driven policy decisions.

1.2 Objectives

General Objectives

To design, develop, and deploy a web-based streamlined employment platform and applicant management platform called STRAM-PESO system using the MERN stack for LMD-PESO Marinduque that automates job posting, that connects job seekers (residents), employers, and administrators for efficient recruitment and talent by the end of the current semester.

Specific Objectives

1. Create a secure login system where users can sign up and log in safely, with different permission levels for job seekers, employers, and admins so everyone can only do what they're supposed to do.
2. Build an API with over 36 endpoints that handles all the important functions like managing users, job posts, applications, messages, and invitations so the app can work smoothly and quickly.
3. Add a real-time chat feature using Socket.IO where employers and job seekers can message each other instantly with message history and typing indicators.
4. Make it easy for employers to post jobs and manage applications from job seekers, including allowing users to upload their resumes and other important documents.
5. Create an admin dashboard that shows useful information like how many users are on the platform, how many jobs have been posted, and tracks employer verification status.
6. Build a profile system where job seekers can list their skills and experience, and employers can add their company information and upload business documents for verification.
7. Deploy the finished application to the cloud so it's available online, working properly, and stays running without major problems.

8. Make sure all information entered by users is checked and validated to prevent errors and keep the platform safe from attacks.
9. Write clear documentation for the API with examples and guides so other developers understand how to use it and other teams know how the system works.
10. Test the whole application to make sure everything works correctly, is secure, and works well on different browsers and devices.

1.3 Review of Related Literature and Systems

Related System 1: JobStreet Philippines

JobStreet Philippines is an online employment marketplace that links job seekers and companies using a consolidated system. It enables users to explore and apply for jobs based on geography, industry, and job type, while businesses can post openings, screen applications, and manage the recruiting process. The platform also employs data-driven matching and suggestions to enhance job relevance and facilitate more efficient recruiting in the Philippines and Asia-Pacific [8].

However, despite its wide adoption, JobStreet primarily focuses on private-sector recruitment and does not fully support government employment services such as PESO integration or localized employment programs, which limits its accessibility for grassroots job seekers. The proposed STRAM-PESO system addresses this limitation by focusing on local government employment facilitation and community-based recruitment services.

Related System 2: Indeed, Job Search Platform

Indeed is a global employment search engine that is commonly used in the Philippines for job postings and recruiting. It allows users to look for job openings by merging listings from several sources, such as corporate websites and employer ads, into a single platform. Job searchers may upload resumes, apply for openings immediately, and narrow their search using parameters such as location, pay, and job type. Indeed also offers job notifications and business reviews, which assist consumers make educated career decisions and speed the application process [9].

Despite its effectiveness, Indeed lacks localized government employment support mechanisms and direct coordination with public employment service offices, making it less suitable for addressing region-specific employment needs such as those handled by PESO offices.

Related System 3: LinkedIn Recruitment Platform

LinkedIn is a professional networking and recruitment website that links job searchers with companies via digital profiles and hiring tools. It enables users to establish profiles, exhibit abilities, search for employment, and apply directly to open positions. LinkedIn Recruiter enables employers to find candidates, screen applicants, manage talent pipelines, and interact via chat tools. The platform also employs AI-powered suggestions and analytics to boost recruiting efficiency and help corporate recruitment globally [10].

Although LinkedIn improves professional networking and digital recruitment, it primarily caters to high-skilled professionals and corporate sectors, leaving many entry-level and local job seekers underserved. STRAM-PESO aims to address this gap by providing a more inclusive employment platform accessible to all job seekers regardless of skill level or location.

Related System 4: Philippines-Jobs.com Recruitment Portal

A specialized online employment site in the Philippines, Philippines-Jobs.com uses a single platform to link companies and job seekers. Employers may publish job adverts and effectively manage recruiting while users can browse and apply for open positions online. Along with career materials that assist users in their job search and employability, the platform also offers a Talent Management System (TMS) that expedites application processing and increases hiring speed. [11].

However, while the platform is effective in facilitating online recruitment, its Talent Management System (TMS) may have limited integration with broader government employment systems and localized public job programs. As a result, it may not fully support coordinated employment initiatives or community-based job matching services handled by Public Employment Service Offices (PESO).

Related System 5: PESO Employment Information System

The PESO Employment Information System (PEIS) in Marinduque is a government-managed platform under PESO that centralizes job seekers' and employers' data to support local job matching and employment services [12].

Its efficiency may be affected by problems like irregular updates and manual data encoding, despite its utility. Because of this, the planned STRAM PESO system is put into place to enhance and modernize PESO services by means of a more efficient and automated applicant management system.

Related System 6: DOLE PhilJobNet

The DOLE PhilJobNet is an online job matching portal developed by the Department of Labor and Employment (DOLE) that connects job seekers and employers nationwide. It provides services such as job search, online application, and labor market information to support employment facilitation in the Philippines [13]

Despite its wide coverage and ability to include local job postings, it mostly functions as a centralized national platform, which may fail to capture highly localized employment demands and PESO-specific community-based recruiting procedures. In light of this, the suggested STRAM PESO system is deployed to improve localized job finding through a more organized, automated, and community-oriented employment platform.

Related System 7: Kalibrr Online Recruitment Platform

The Kalibrr Online Recruitment Platform is a digital hiring system that connects job seekers and employers through an AI-based job matching and recruitment platform, allowing online applications and automated screening [14].

Despite its innovative recruitment tools, Kalibrr is mainly designed for private-sector recruitment, which limits its effectiveness for public employment service operations.

Related System 8: Foundit (Monster) Talent Platform

The Foundit (Monster) Talent Platform is a global recruiting system that links job seekers and businesses with a technologically advanced job site that provides tailored job matching, resume search, and talent management solutions. It has grown into an end-to-end platform that delivers personalized job suggestions and recruiting tools for both seekers and businesses. [15].

However, global recruitment platforms like Foundit do not specifically address localized employment needs and government-managed recruitment programs.

Related System 9: Manila Recruitment Online Recruitment Platform

The Manila Recruitment Online Recruitment Platform is a digital recruiting service in the Philippines that links companies with job seekers using executive search, technical recruitment, and talent sourcing solutions. It offers complete recruiting services, such as applicant sourcing, screening, and placement, for a variety of industries, including IT, executive, and professional jobs [16].

Nevertheless, providing specialist recruiting and professional personnel sourcing services to corporate clients, the platform focuses mostly on private-sector and high-level hiring, limiting its support for mass local employment and community-based job matching. In this case, the planned STRAM PESO system is deployed to create a more inclusive, localized, and government-integrated employment platform that better meets public job placement demands.

Related System 10: Technical Education and Skills Development Authority (TESDA)

The Technical Education and Skills Development Authority (TESDA) is a Philippine government organization responsible for managing and supervising technical-vocational education and training. It offers skill development programs, competence certification, and training opportunities to improve employability and workforce competitiveness in the country [17].

TESDA, although offering a wide range of training programs and employment-related services, concentrates on skill development rather than direct job-finding systems, and it may have limited connection with localized employment platforms like PESO. In light of this, the planned STRAM PESO system is designed to give a more organized, localized, and government-integrated employment-finding platform.

Related System 11: Careerjet Online Job Search Engine

The Careerjet Online Job Search Engine is a global job aggregator that combines job postings from a variety of recruiting websites, job boards, and firm career pages into a single searchable database. It allows users to look for career possibilities across many nations, including the Philippines, by sending them to the original job posts for application [18].

Considering its extensive job coverage and aggregation of postings from many sources, Careerjet primarily serves as a job search engine and does not offer comprehensive recruiting capabilities such as application tracking or interface with government systems such as PESO. In light of this, the planned STRAM PESO system is being constructed to provide a more localized, organized, and government-supported job-finding platform.

Related System 12: OnlineJobs.ph E-Recruitment Platform

OnlineJobs.ph is an online recruiting platform that links companies with Filipino professionals, mostly virtual assistants and remote workers. It allows companies to hire, communicate, and manage employees directly, which supports flexible employment arrangements including part-time, full-time, and freelance work. [19].

However, despite its enormous database of Filipino workers and fast remote recruiting mechanism, the platform is mostly focused on private freelancing and outsourced activities, with no interface with government employment systems like as PESO, LGU, and others. By considering this, the planned STRAM PESO system is constructed to give a more localized, organized, and government-integrated job search platform.

Related System 13: Bossjob AI Recruitment Platform

The Bossjob AI Recruitment Platform is a Philippine online job platform that links employment seekers and businesses using a chat-based hiring system. It enables users to apply for jobs, contact directly with recruiters, and receive AI-powered job suggestions to accelerate the recruiting process [20].

Despite delivering rapid AI-based job matching and direct recruiter chat tools, Bossjob's primary concentration is on private-sector positions such as IT, BPO, sales, and startup-related professions, which limits its coverage of larger community-based and government-assisted employment initiatives. In regard to this, the planned STRAM PESO system is designed to give a more localized, organized, and government-integrated job search platform that meets a wider range of employment demands.

Related System 14: Glassdoor Job and Company Review Platform

The Glassdoor platform is an online employment and company review website where job searchers may find job postings, salary statistics, and employee-generated company evaluations. It also allows firms to publish job openings and manage corporate branding in order to attract competent applicants. [21].

While Glassdoor offers useful information such as pay comparisons, company reviews, and employee feedback, its primary focus is on employer transparency and job seeking, rather than true end-to-end recruiting or applicant handling. In regard to this, the planned STRAM PESO system is constructed to provide a more centralized, organized, and government-integrated job searching platform focused on local-based employment services

Related System 15: AI-Based Fair Recruitment Frameworks

The Jobsora Online Job Search Platform is a website that gathers job postings from many sources and displays them in a consolidated search interface. It allows users to view job vacancies based on location, including Quezon City, and offers criteria such as job title, category, and industry to increase job search accessibility [21].

Despite its extensive job listings and ease of aggregating vacancies from various employers, Jobsora primarily serves as a job search engine rather than a structured recruitment or government employment system, and it does not support full applicant tracking or integration with PESO and other local government employment services.

1.4 Methodology & Solution

This project will employ the Agile methodology with 2-week sprints, allowing iterative feedback from LMD-PESO stakeholders. The solution will be built using the MERN stack (MongoDB, Express.js, React.js, Node.js).

System Architecture:

- Frontend (React.js): A responsive single-page application (SPA) with separate dashboards for Job Seekers, Employers, PESO Staff, and Admin. UI will use CSS for rapid, accessible design.
- Backend API (Express.js/Node.js): RESTful API endpoints for authentication (JWT + bcryptjs), job CRUD, application submission, resume upload, and report generation (PDF/Excel using jsPDF or SheetJS).
- Database (MongoDB): Four primary collections: Users (with role field), Jobs, Applications, and Documents.

Solution Workflow (Backend to Frontend):

1. Authentication: User logs in → credentials sent to `/api/auth/login` → JWT token returned → token stored in HTTP-only cookie or local Storage → frontend attaches token to subsequent requests.
2. Job Posting (Employer/PESO Staff): Form data submitted → validated → saved to Jobs collection → real-time notification sent to matched job seekers

Testing Strategy:

- Integration Testing (Supertest): Test API endpoints (e.g., `POST /api/jobs` returns 201, `GET /api/reports` returns PDF).
- End-to-End Testing (Cypress): Simulate user flows (e.g., "Job seeker registers → searches for job → applies → PESO staff approves application").
- User Acceptance Testing (UAT): Conducted with 10 real LMD-PESO staff, 20 job seekers, and 5 employers from Marinduque.

1.5 Target Users / Beneficiaries

The STRAM-PESO system encompasses the digital transformation of the manual employment facilitation processes currently used by the LMD-PESO Marinduque. It is designed as a centralized, web-based hub serving three primary user groups:

- LMD-PESO Administrator and Staff
 - Manage and verify employer accounts and job vacancies.
 - Monitor applicant data and digital files remotely via a secure database.
 - Generate one-click reports for provincial or national government submissions.
 - Broadcast notifications regarding programs, services, and job openings.
- Applicants
 - Create digital profiles that serve as automated resumes.
 - Upload and manage certifications, clearances, and resumes in a digitized repository.
 - Apply for jobs online and track application progress through a dedicated dashboard.
- Employers and Recruitment Agencies
 - Post job vacancies once verified by the LMD-PESO.
 - Access a pool of pre-screened candidates based on specific requirements.

1.6 Significance of the Study

The researchers intend to provide a platform named STRAM-PESO: An Online Employment and Applicant Management System for Public Employment Service Office (LMD-PESO) that will enhance the employment facilitation system for the constituents of Marinduque. This web application will aim to improve work efficiency by automating the current methods that still rely on traditional means.

The first beneficiary of this system is the organization itself. It will be easier for the organization to monitor the files of the clients such as personal information. The files of the clients will be much easier to locate since it will be stored in a secure database. Unlike the present system, files are stored in cabinets or drawers and there is a vast chance of misplacement or loss of documents. Another benefit for the organization is that they can access the files remotely as long as they have access to the internet. For the job posting, the post will spread to the masses and may gather a larger scale of audience faster since it will be posted online.

The second beneficiaries of the system are PESO clients, none-other than the job seekers. They will have an opportunity to get a job that is suitable for them with the help of a more accurate job matching system. It will be easier for them to apply because just by creating their profile as a job seeker, this will also serve as their resume to be able to apply for a job. There will also be an available job posting on the system. This will help the jobseekers because they will no longer have to spend and go out to find a job that suits their skills and knowledge.

The final beneficiaries of the system are companies, recruiting agencies or in other words, employers affiliated with the LMD-PESO. It will be easier for them to find employees because they will have access to the system. They can post job vacancies as long as they are verified by PESO.

CHAPTER 2: MINIMUM VIABLE PRODUCT (MVP) DOCUMENTATION

2.1 System Architecture

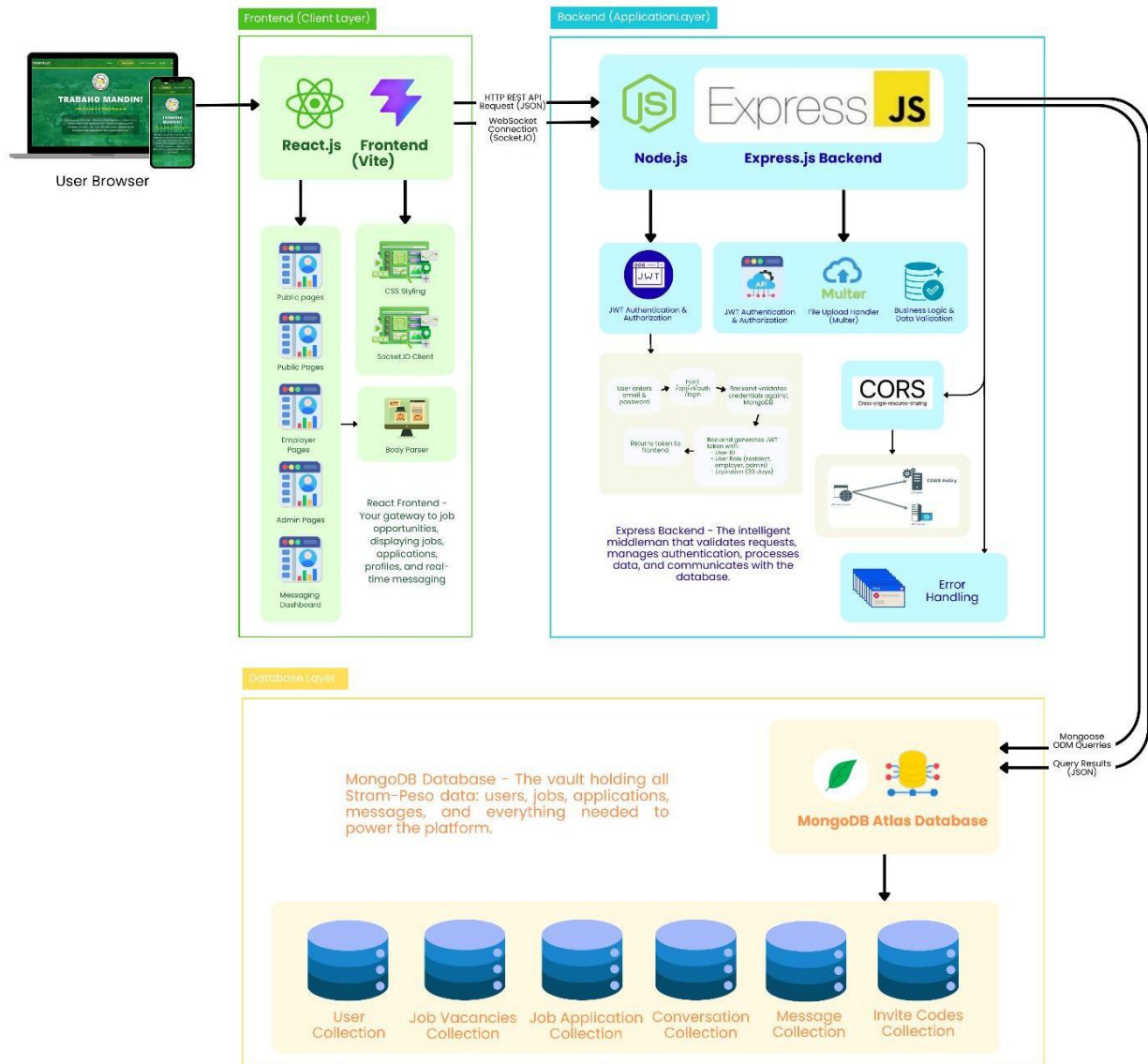


Figure 1: Stram Peso System Architecture

2.2 Tech Stack & Tools

Frontend (Client-Side)

- React.js - UI library for building interactive user interfaces
- Vite - Fast build tool for development and production bundling
- React Router - Client-side routing for navigation between pages
- CSS Frameworks - Custom CSS stylesheets (App.css, dashboard.css, navbar.css, etc.)
- Socket.IO Client - For real-time messaging and notifications
- Axios/Fetch API - HTTP client for API communication

Backend (Server-Side)

- Node.js - JavaScript runtime environment for server
- Express.js - Minimalist web framework for building REST API
- Middleware - CORS, body-parser, multer for file uploads
- Socket.IO - Real-time, bidirectional communication between client and server
- Multer - File upload handling in Node.js
- JWT (jsonwebtoken) - Token-based authentication
- bcryptjs - Password hashing and encryption

Database

- MongoDB Atlas - Cloud-hosted NoSQL database
- Mongoose - ODM (Object Data Modeling) for MongoDB
- Connection String - mongodb+srv://user:password@cluster.mongodb.net/dbname

Testing & Quality

- Postman - Manual API testing and testing collections
- Jest - Unit testing framework (optional for backend)
- Supertest - HTTP assertion library for testing endpoints

Deployment & Hosting

- Vercel - Frontend deployment (React app hosting)
- MongoDB Atlas - Database hosting (cloud)
- Render/Heroku - Backend deployment option (Node.js server)
- Environment Variables - [.env](#) file for sensitive data (JWT_SECRET, DB_URI)

Development Tools

- Git/GitHub - Version control and collaboration
- VSCode - Code editor
- NPM - Package manager for Node.js dependencies
- Dotenv - Environment variable management

2.3 Database Schema (Data Modeling)

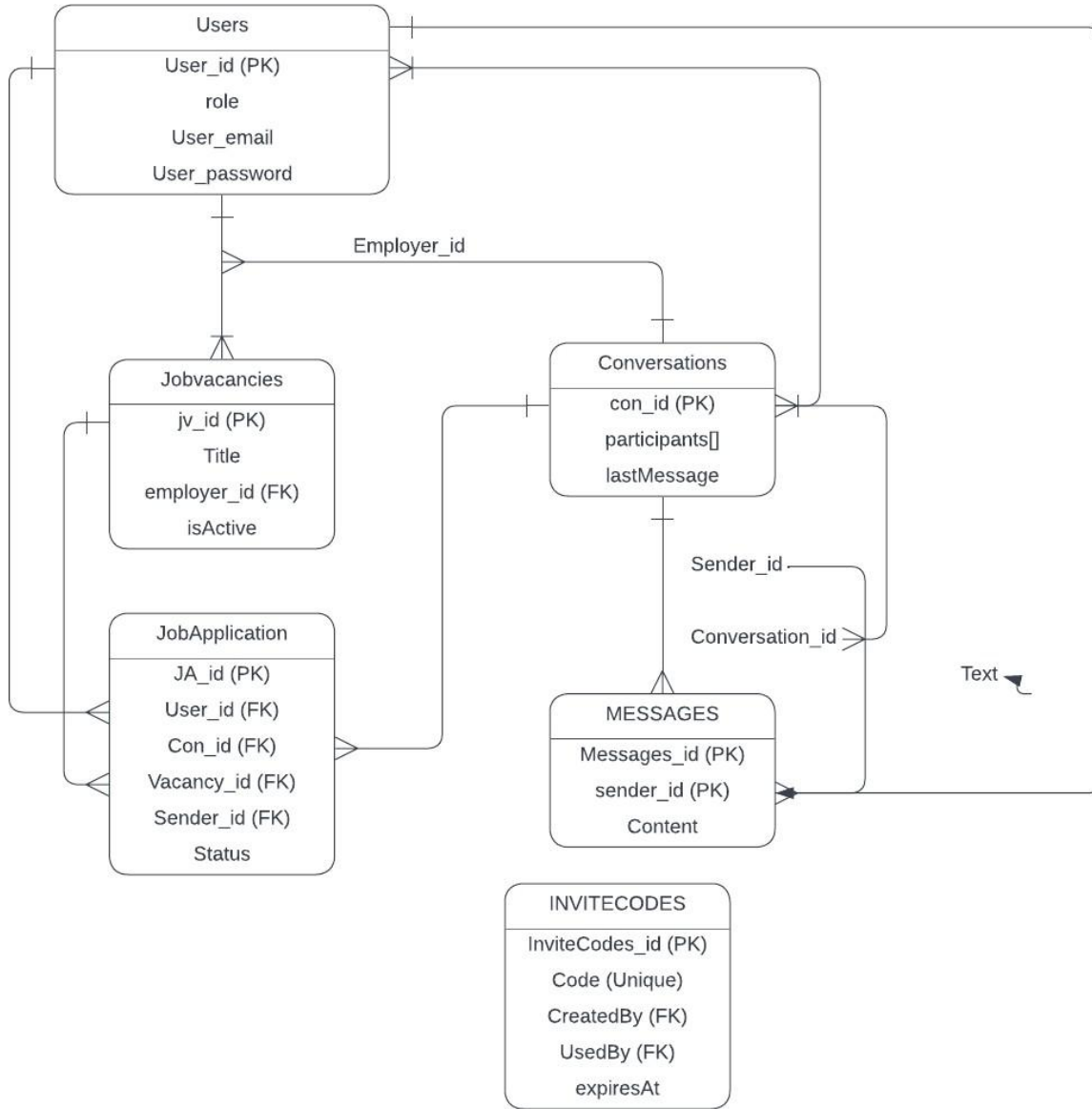


Figure 2: Database Schema

CHAPTER 3: REST API DESIGN DOCUMENTATION

3.1 Overview & Intent

Base URL: `https://your-backend-domain.vercel.app/api/v1`

Purpose: Stram-Peso is a comprehensive employment and community platform that manages user authentication, job postings and applications, real-time messaging between job seekers and employers, user profile management, announcements, and administrative analytics. The platform connects residents (job seekers), employers (recruiters), and administrators through a secure role-based access system.

3.2 Authentication & Authorization

Strategy: JSON Web Tokens (JWT) passed via the Authorization: Bearer <token> header.

Roles:

- admin: Full system access - can manage users, view analytics, delete reports, promote users to employers, generate invite codes
- employer: Can create and manage job postings, view applications, update application status, post announcements, message applicants, manage company profile
- resident: Can view jobs, apply to jobs, view announcements, message employers, manage personal profile, update application status

3.3 Endpoint Documentation

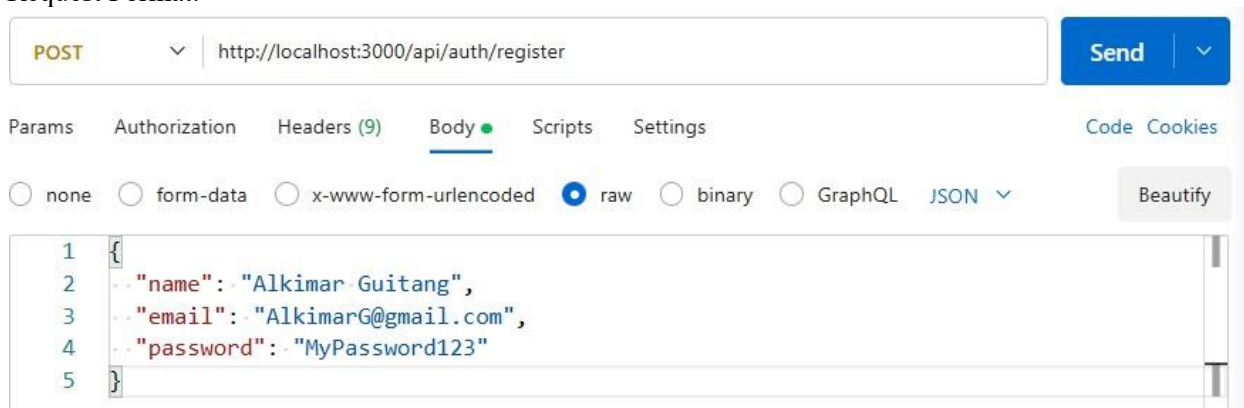
AUTHENTICATION ENDPOINTS

1. Register User

Method & URL: POST /auth/register

Authorization Required: No

Request Format:

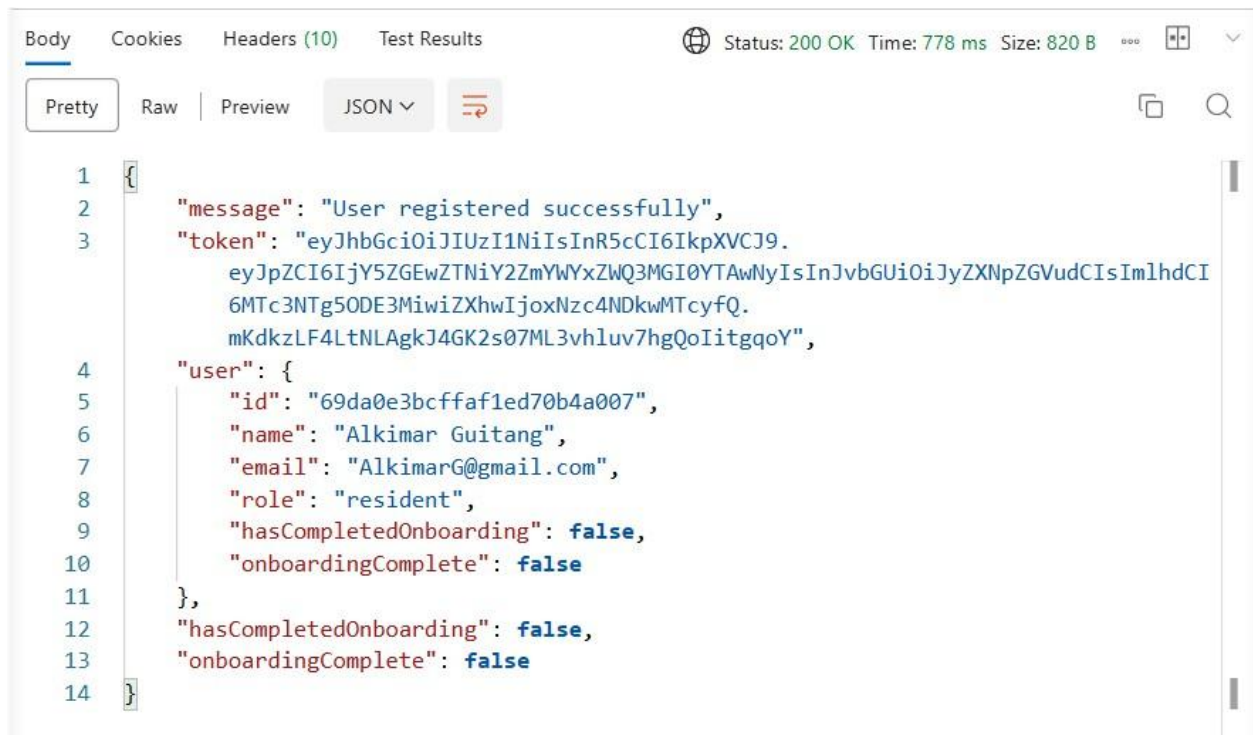


The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** `http://localhost:3000/api/auth/register`
- Body:** A JSON object with the following fields:

```
{  1  {  2    "name": "Alkimar Guitang",  3    "email": "AlkimarG@gmail.com",  4    "password": "MyPassword123"  5  }
```
- Format:** JSON
- Buttons:** Send, Beautify

Success Response (200 OK):



```
1 {
2   "message": "User registered successfully",
3   "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5ZGEwZTNiY2ZmYWYxZWQ3MGI0YTAwNyIsInJvbGUiOiJyZXNpZGVudCI6ImIhdCI6MTc3NTg5ODE3MiwiaXhwIjoxNzc4NDkwMTcyfQ.mKdkzLF4LtNLAgkJ4GK2s07ML3vhluv7hgQoIitgqoY",
4   "user": {
5     "id": "69da0e3bcffaf1ed70b4a007",
6     "name": "Alkimar Guitang",
7     "email": "AlkimarG@gmail.com",
8     "role": "resident",
9     "hasCompletedOnboarding": false,
10    "onboardingComplete": false
11  },
12  "hasCompletedOnboarding": false,
13  "onboardingComplete": false
14 }
```

Error Response (400 Bad Request):



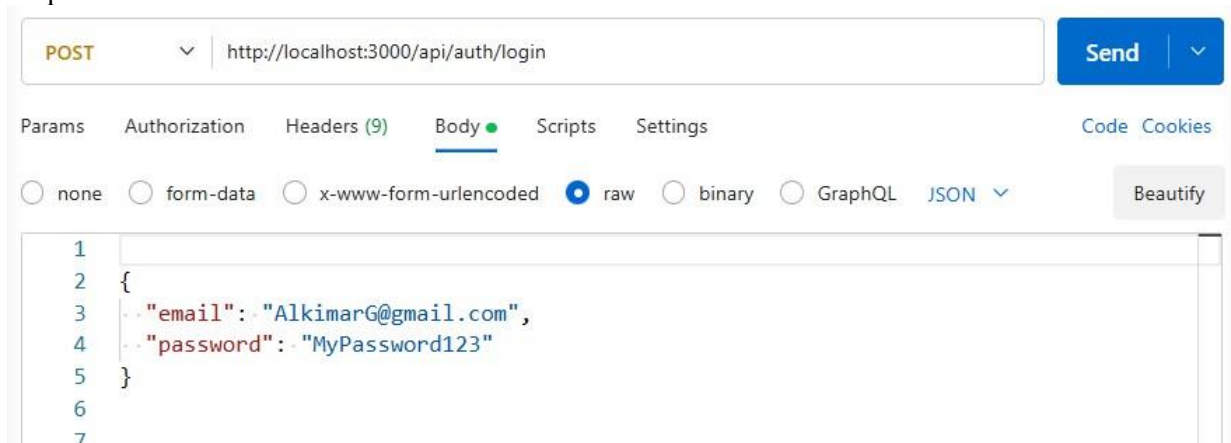
```
1 {
2   "message": "Email already exists"
3 }
```

2. User Login

Method & URL: POST /auth/login

Authorization Required: No

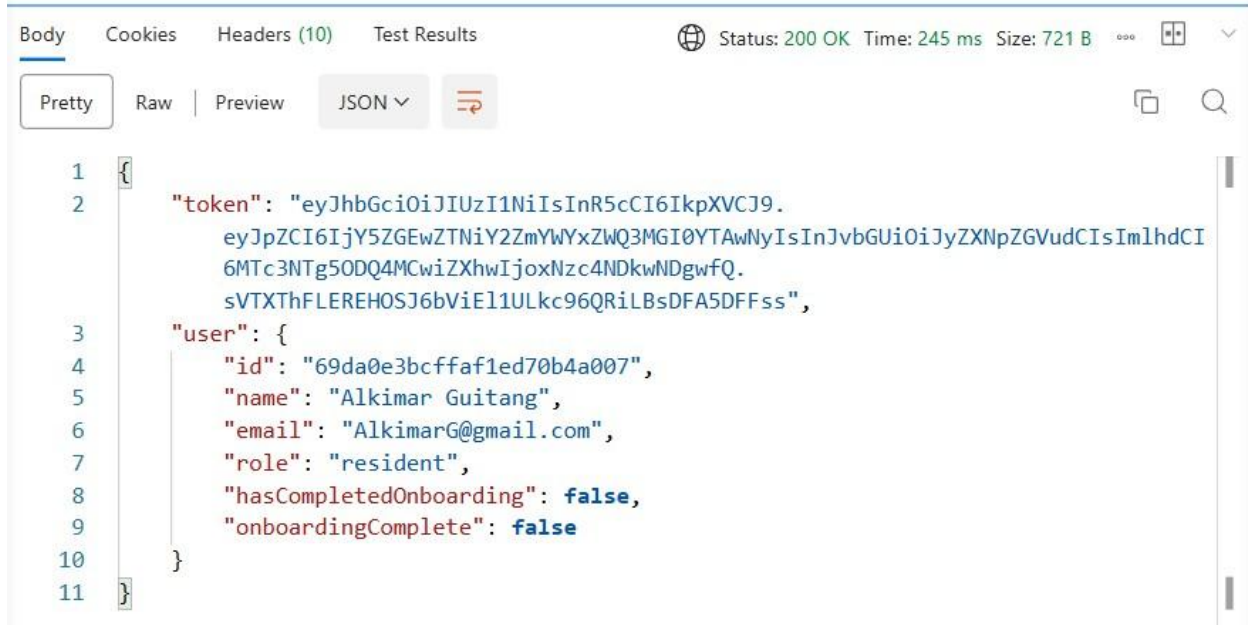
Request Format:



```
POST http://localhost:3000/api/auth/login

1 {
2   "email": "AlkimarG@gmail.com",
3   "password": "MyPassword123"
4 }
```

Success Response (200 OK):

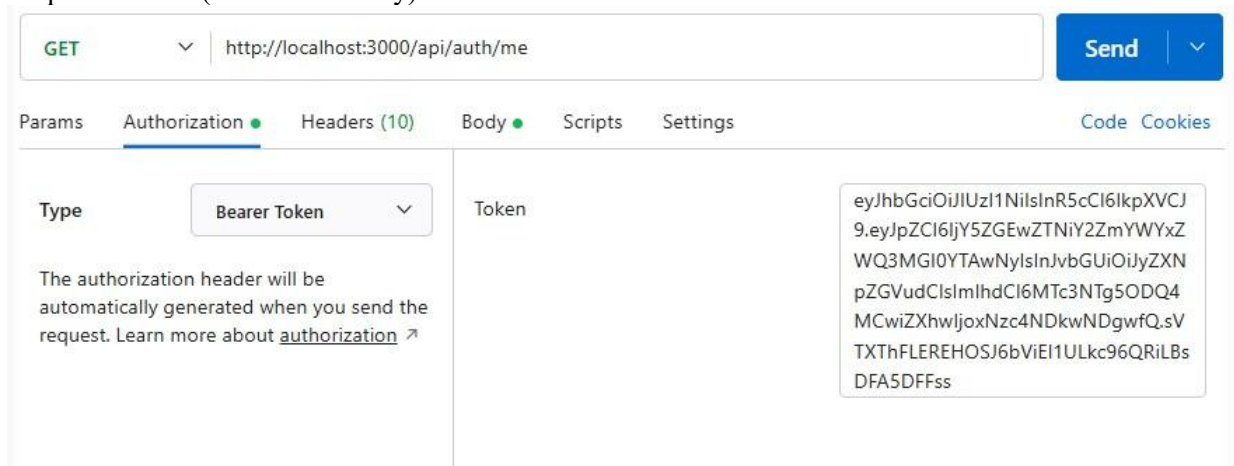


3. Get Current User

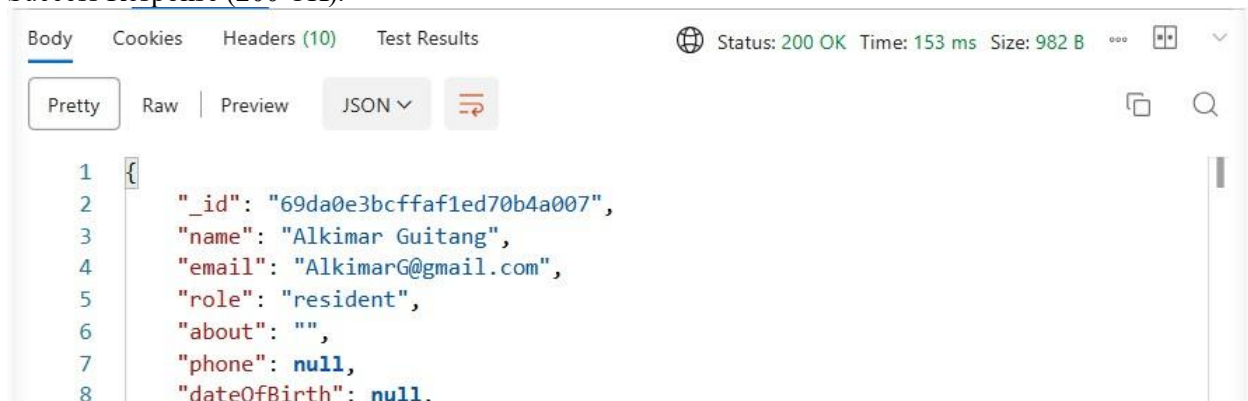
Method & URL: GET /auth/me

Authorization Required: Yes (All Roles)

Request Format (Headers & Body):



Success Response (200 OK):



```

8     "dateOfBirth": null,
9     "gender": null,
10    "address": null,
11    "skills": [],
12    "desiredJobTitle": null,
13    "educationalAttainment": null,
14    "workExperience": null,
15    "availabilityStatus": null,
16    "companyName": "",
17    "industry": "",
18    "companySize": "",
19    "website": "",
20    "companyDescription": "",
21    "businessAddress": "",
22    "verificationStatus": "unverified",
23    "businessPermitUrl": null,
24    "registrationDocUrl": null,
25    "resumeFile": null,
26    "validIdFile": null,
27    "isActive": true,
28    "hasCompletedOnboarding": false,
29    "onboardingComplete": false,
30    "createdAt": "2026-04-11T09:02:51.846Z",
31    "__v": 0
32 }

```

Error Response (403 Not Found):

Body Cookies Headers (10) Test Results 🌐 Status: 403 Forbidden Time: 14 ms Size: 375 B ...

Pretty Raw Preview JSON 

```

1 {
2   "message": "Invalid token"
3 }

```

4. Update User Profile

Method & URL: PUT /auth/profile

Authorization Required: Yes (All Roles)

Request Format: (Multipart form data)

PUT ⌵ http://localhost:3000/api/auth/profile Send ⌵

Params Authorization ● Headers (10) Body ● Scripts Settings Code Cookies

Type Bearer Token ⌵

The authorization header will be automatically generated when you send the request. [Learn more about authorization](#) ↗

Token

```

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5SGEwZTNiY2ZmYWYxZWQ3MG10YTAwNyIsInJvbGUiOiJyZXNpZGVudCIsImhhdCI6MTc3NTg5ODQ4MCwiZXhwIjoxNzc4NDkwNDgwfwQ.sVTXThFLEREHOSJ6bViE11ULkc96QRiLBsDFA5DFFss

```

Params Authorization Headers (10) **Body** Scripts Settings Code Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON

```
1 |
2 {
3   "email": "AlkimarG@gmail.com",
4   "phone": "+1234567890",
5   "dateOfBirth": "2004-01-08",
6   "gender": "Male",
7   "address": "Tamayo, Santa Cruz, Marinduque",
8   "skills": "JavaScript,React,Node.js",
9   "desiredJobTitle": "Senior Developer",
10  "educationalAttainment": "College Graduate",
11  "workExperience": "3-5 years",
12  "availabilityStatus": "Actively Looking"
13 }
14
```

Success Response (200 OK):

Body Cookies Headers (10) Test Results  Status: 200 OK Time: 170 ms Size: 1.13 KB  

Pretty Raw Preview JSON 

```
1 {
2   "message": "Profile updated",
3   "user": {
4     "_id": "69da0e3bcffaf1ed70b4a007",
5     "name": "Alkimar Guitang",
6     "email": "AlkimarG@gmail.com",
7     "role": "resident",
8     "about": "",
9     "phone": "+1234567890",
10    "dateOfBirth": "2004-01-08T00:00:00.000Z",
11    "gender": "Male",
12    "address": "Tamayo, Santa Cruz, Marinduque",
13    "skills": [
14      "JavaScript",
15      "React",
16      "Node.js"
17    ],
18    "desiredJobTitle": "Senior Developer",
19    "educationalAttainment": "College Graduate",
20    "workExperience": "3-5 years",
21    "availabilityStatus": "Actively Looking",
22    "companyName": "",
23    "industry": "",
24    "companySize": "",
25    "website": ""
26  }
27 }
```

5. Delete Account

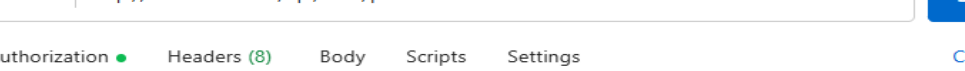
Method & URL: DELETE /auth/profile

Authorization Required: Yes (All Roles)

Request Format (Headers & Body):

[illegible]

Success Response (200 OK):



The screenshot shows a REST client interface with the following details:

- Method:** DELETE
- URL:** http://localhost:3000/api/auth/profile
- Buttons:** Send
- Tabs:** Params, Authorization (selected), Headers (8), Body, Scripts, Settings
- Response Status:** 200 OK, Time: 194 ms, Size: 383 B
- Response Body:**

```
{
  "message": "Account successfully deleted"
}
```

EMPLOYER ENDPOINTS

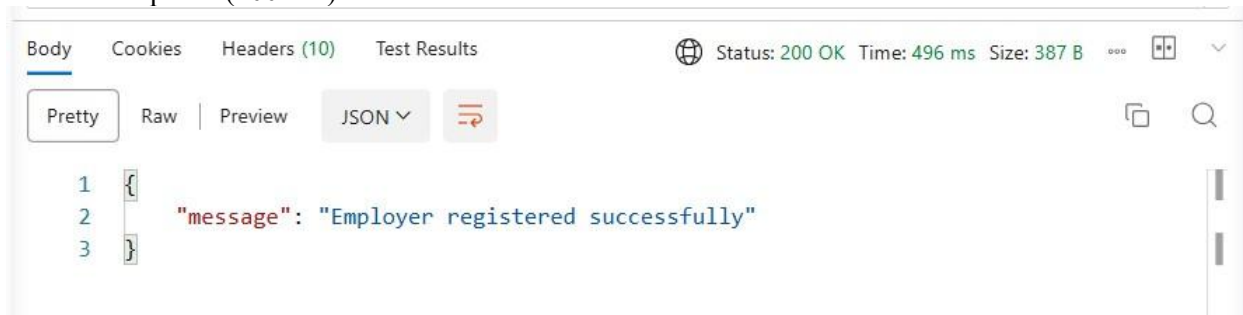
1. Register as Employer

Method & URL: POST /auth/register/employer

Authorization Required: No

Request Format:

Success Response (200 OK):

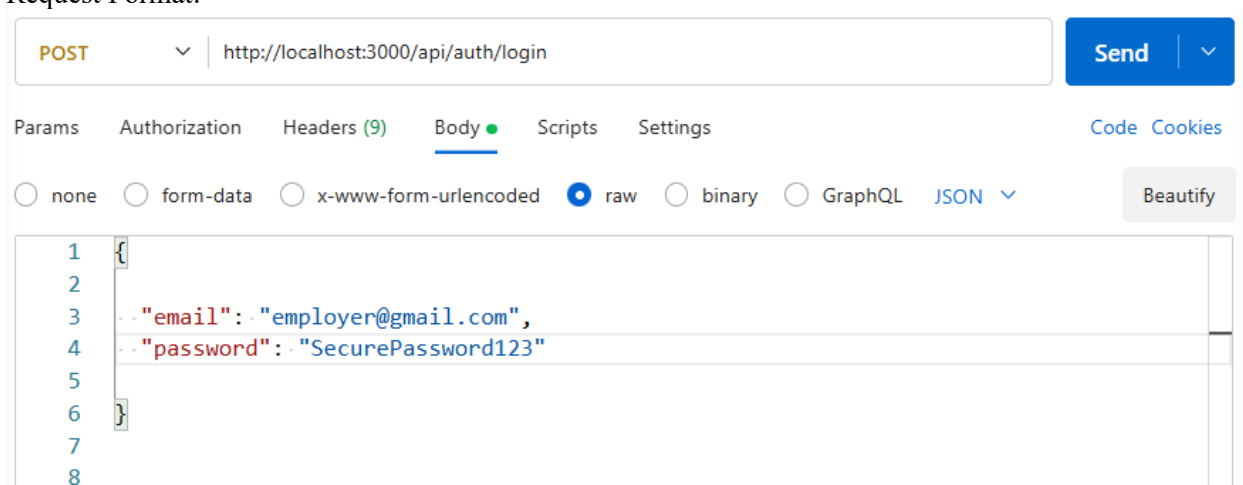


2. Employer Login

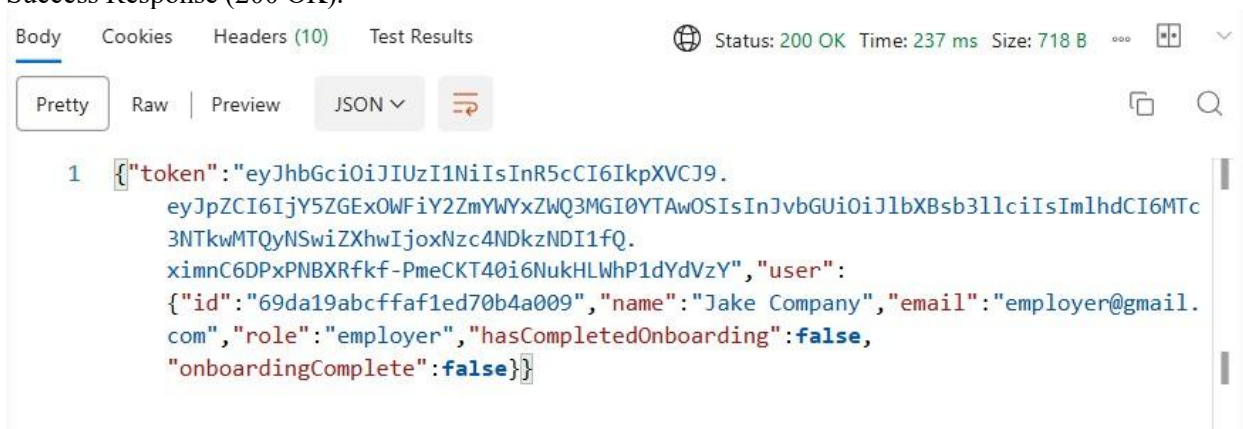
Method & URL: POST /auth/login

Authorization Required: No

Request Format:



Success Response (200 OK):



JOB ENDPOINTS

1. Create Job Posting

Method & URL: POST /jobs

Authorization Required: Yes (Employer only)

POST

http://localhost:3000/api/auth/jobs

Send

ParamsAuthorizationHeaders (9)BodyScriptsSettingsCodeCookies

Type

Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#)

Token

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5ZGExOWFiY2ZmYWYxZlQ3MGI0YTAwOSIsInJvbnUiOiJlYXBsYXN1bGciLCJhdCI6MTc3NTkwMTQyNSwiZXhwIjoxNzQ0NDkzNDI1fQ.ximn6DpXPNBXRfkf-PmeCKT40i6NukHLWhP1dYdVzY"

Request Format:

POST

http://localhost:3000/api/jobs

Send

ParamsAuthorizationHeaders (10)BodyScriptsSettingsCodeCookies

☐ none

☐ form-data

☐ x-www-form-urlencoded

☒ raw

☐ binary

☐ GraphQL

JSON

Beautify

```
1 {
2   "title": "Senior React Developer",
3   "description": "Looking for experienced React developer with 3+ years
  experience",
4   "location": "Manila, Philippines",
5   "salary": "₱80,000 - ₱120,000/month",
6   "requirements": "3+ years React experience, TypeScript, REST APIs",
7   "jobType": "Full-time",
8   "slots": 2
9 }
```

Success Response (200 OK):

BodyCookiesHeaders (10)Test Results

Status: 200 OK Time: 216 ms Size: 816 B

PrettyRawPreviewJSON

```
1 {"message": "Job posted successfully", "job": {"title": "Senior React Developer",
  "description": "Looking for experienced React developer with 3+ years
  experience", "location": "Manila, Philippines", "salary": "₱80,000 - ₱120,000/
  month", "requirements": "", "jobType": "Full-time", "slots": 1,
  "employer": "69da19abcffaf1ed70b4a009", "isActive": true, "status": "active",
  "_id": "69da1bd5cfffaf1ed70b4a00a", "createdAt": "2026-04-11T10:00:53.120Z",
  "updatedAt": "2026-04-11T10:00:53.120Z", "__v": 0}}
```

2. Get All Jobs

Method & URL: GET /jobs

Authorization Required: Yes (All Roles)

Query Parameters:

?limit=20&page=1&location=Manila&jobType=Full-time&sort=-createdAt

GET

http://localhost:3000/api/jobs?limit=20&page=1&location=Manila&jobType=Full-time&sort...

Send

Params

Authorization

Headers (8)

Body

Scripts

Settings

Code

Cookies

Type

Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#)

Token

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5ZGEwOFYyZWmYWYxZWQ3MGI0YTAwOSIsInJvbnUiOiJlbnB3b3llciIsImIhdCI6MTc3NTkwMTQyNSwiZXhwIjoxNzc4NDkzNDI1fQ.ximnC6DPxPNBXRfkf-PmeCKT40i6NukHLWhP1dYdVzY

Success Response (200 OK):

Body

Cookies

Headers (10)

Test Results

Status: 200 OK Time: 367 ms Size: 1004 B

Pretty

Raw

Preview

JSON

```
1 [
2   {
3     "_id": "69da1bd5cffaf1ed70b4a00a",
4     "title": "Senior React Developer",
5     "description": "Looking for experienced React developer with 3+ years
6       experience",
7     "location": "Manila, Philippines",
8     "salary": "₱80,000 - ₱120,000/month",
9     "requirements": "",
10    "jobType": "Full-time",
11    "slots": 1,
12    "employer": {
13      "_id": "69da19abcfaf1ed70b4a009",
14      "name": "Jake Company",
15      "email": "employer@gmail.com",
16      "role": "employer",
17      "phone": null,
18      "companyName": "",
19      "industry": "",
20      "companySize": "",
21      "website": "",
22      "companyDescription": "",
23      "businessAddress": "",
24      "verificationStatus": "unverified"
25    },
26    "isActive": true,
27    "status": "active",
28    "createdAt": "2026-04-11T10:00:53.120Z",
29    "updatedAt": "2026-04-11T10:00:53.120Z",
30    "__v": 0
31  }
```

Authorization Required: Yes (All Roles)

```
Body    Cookies    Headers    Test Results    Status: 200 OK    Time: 256 ms    Size: 1002 B
Pretty  Raw    Preview    JSON
1 [{"_id":"69da1bd5cffaf1ed70b4a00a","title":"Senior React Developer",
  "description":"Looking for experienced React developer with 3+ years experience",
  "location":"Manila, Philippines","salary":"₱80,000 - ₱120,000/month",
  "requirements":"","jobType":"Full-time","slots":1,"employer":
  {"_id":"69da19abcfaf1ed70b4a009","name":"Jake Company","email":"employer@gmail.
  com","role":"employer","phone":null,"companyName":"","industry":"","
  "companySize":"","website":"","companyDescription":"","businessAddress":"","
  "verificationStatus":"unverified"},"isActive":true,"status":"active",
  "createdAt":"2026-04-11T10:00:53.120Z","updatedAt":"2026-04-11T10:00:53.120Z",
  "__v":0}]
```

Authorization Required: Yes (Employer - Owner only)

PUT

▼

http://localhost:3000/api/jobs/69da20e9cfff1ed70b4a00c

Send

▼

Params

Authorization

Headers (10)

Body

Scripts

Settings

Code

Cookies

Type

Bearer Token

▼

Token

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5ZGE5OWFiY2ZmYWYxZWQ3MGI0YTAwOSIsInJvbGUiOiJlbXBs3IlciIsImIhdCI6MTc3NTkwMTQyNSwiZXhwIjoxNzc4NDkzNDI1fQ.ximnC6DPxPNBXRfkf-PmeCKT40i6NukHLWhP1dYdVzY

```
1 {
2   "title": "Lead Developer",
3   "description": "Looking for experienced React developer with 3+ years experience",
4   "location": "Manila, Philippines",
5   "salary": "₱80,000 - ₱120,000/month",
6   "requirements": "3+ years React experience, TypeScript, REST APIs",
7   "jobType": "Full-time",
8   "slots": 5
9 }
```

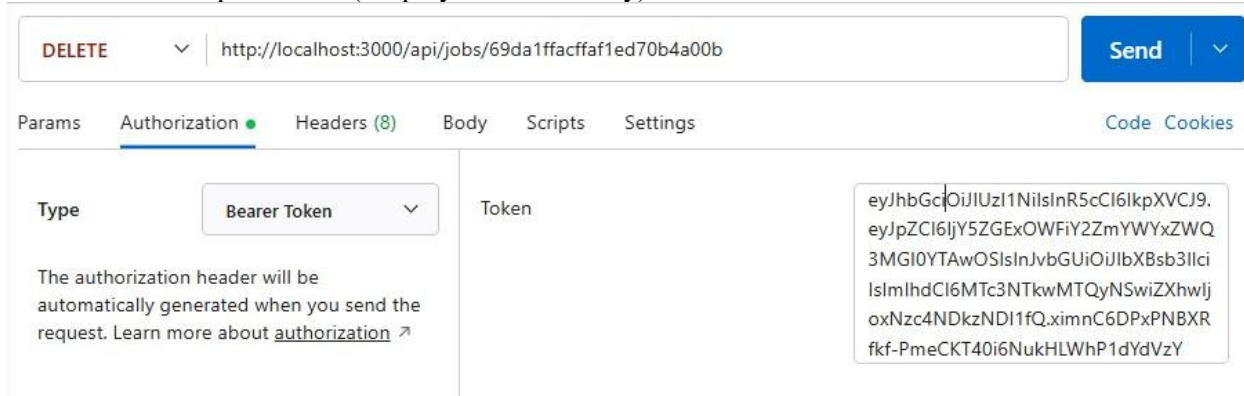
Success Response (200 OK):



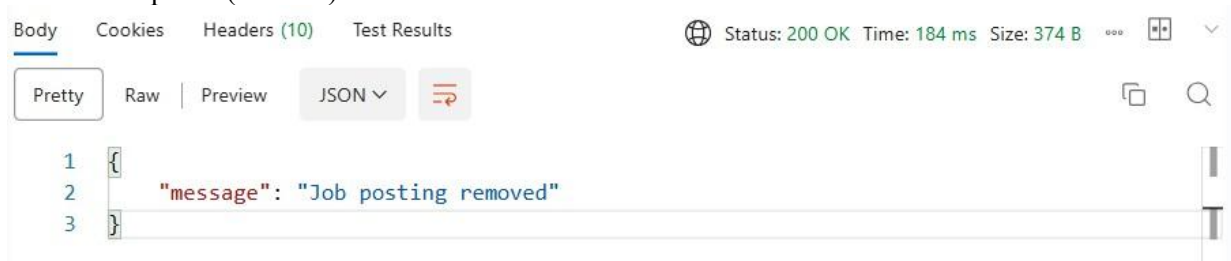
5. Delete Job

Method & URL: DELETE /jobs/:id

Authorization Required: Yes (Employer - Owner only)



Success Response (200 OK):



JOB APPLICATION ENDPOINTS

1. Apply for Job

Method & URL: POST /jobs/:id/apply

Authorization Required: Yes (Resident only)

The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** http://localhost:3000/api/jobs/69da20e9cffaf1ed70b4a00c/apply
- Buttons:** Send, Code, Cookies
- Tabs:** Params, Authorization (selected), Headers (10), Body, Scripts, Settings
- Authorization:** Type: Bearer Token. Token: eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5ZGEwZTNiY2ZmYWYxZWQ3MGI0YTAwNyIsInJvbGUiOiJyZXNpZGVudCIsImIhdCI6MTc3NTkwMzM3NywiZXhwIjoxNzc4NDk1Mzc3fQ.XKu4wqxI9Jg4hkwwOcgIWHHwVLOfkrhwW-P1UPd
- Text:** The authorization header will be automatically generated when you send the request. Learn more about [authorization](#)

Request Format: (Multipart form data)

The screenshot shows a REST client interface with the following details:

- Tabs:** Params, Authorization, Headers (10), Body (selected), Scripts, Settings
- Buttons:** Code, Cookies, Beautify
- Request Format:** none, form-data, x-www-form-urlencoded, raw (selected), binary, GraphQL, JSON
- Body:**

```
1 {
2   "coverLetter": "I am very interested in this position because believe i have
   the experience enough in this field that I am a great developer"
3 }
```

File uploads: resume (Optional)

Success Response (200 OK):

The screenshot shows a REST client interface with the following details:

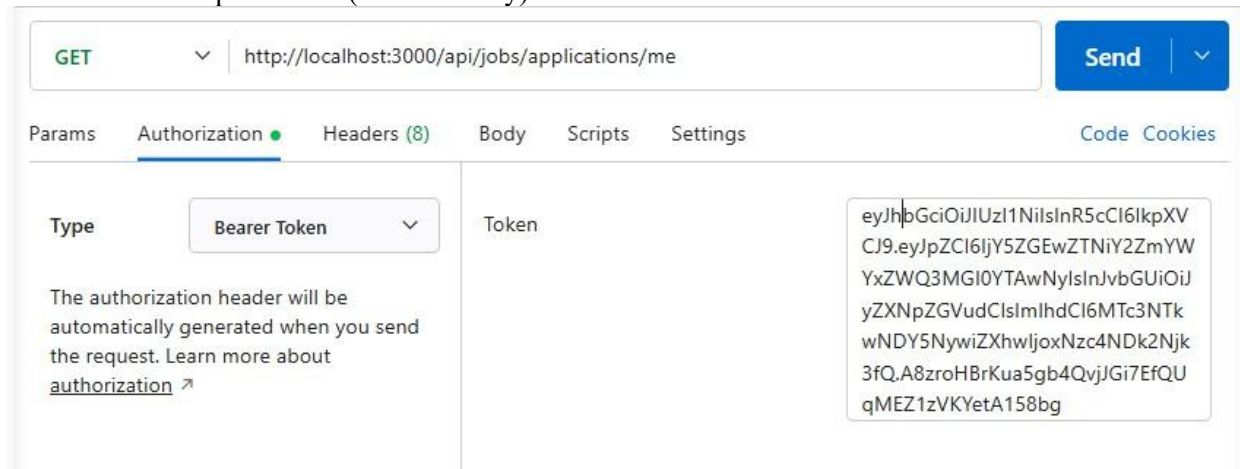
- Tabs:** Body (selected), Cookies, Headers (10), Test Results
- Status:** 200 OK, Time: 230 ms, Size: 803 B
- Buttons:** Pretty (selected), Raw, Preview, JSON, Copy, Search
- Body:**

```
1 {"message": "Application submitted successfully", "application":
  {"applicant": "69da0e3bcffaf1ed70b4a007",
   "vacancy": "69da20e9cffaf1ed70b4a00c", "coverLetter": "I am very interested
   in this position because believe i have the experience enough in this
   field that I am a great developer", "status": "pending", "employerNote": "",
   "statusUpdatedAt": null, "_id": "69da2412cffaf1ed70b4a00d",
   "appliedAt": "2026-04-11T10:36:02.424Z", "createdAt": "2026-04-11T10:36:02.
   430Z", "__v": 0}}}
```

2. Get My Applications

Method & URL: GET /jobs/applications/me

Authorization Required: Yes (Resident only)



GET Send

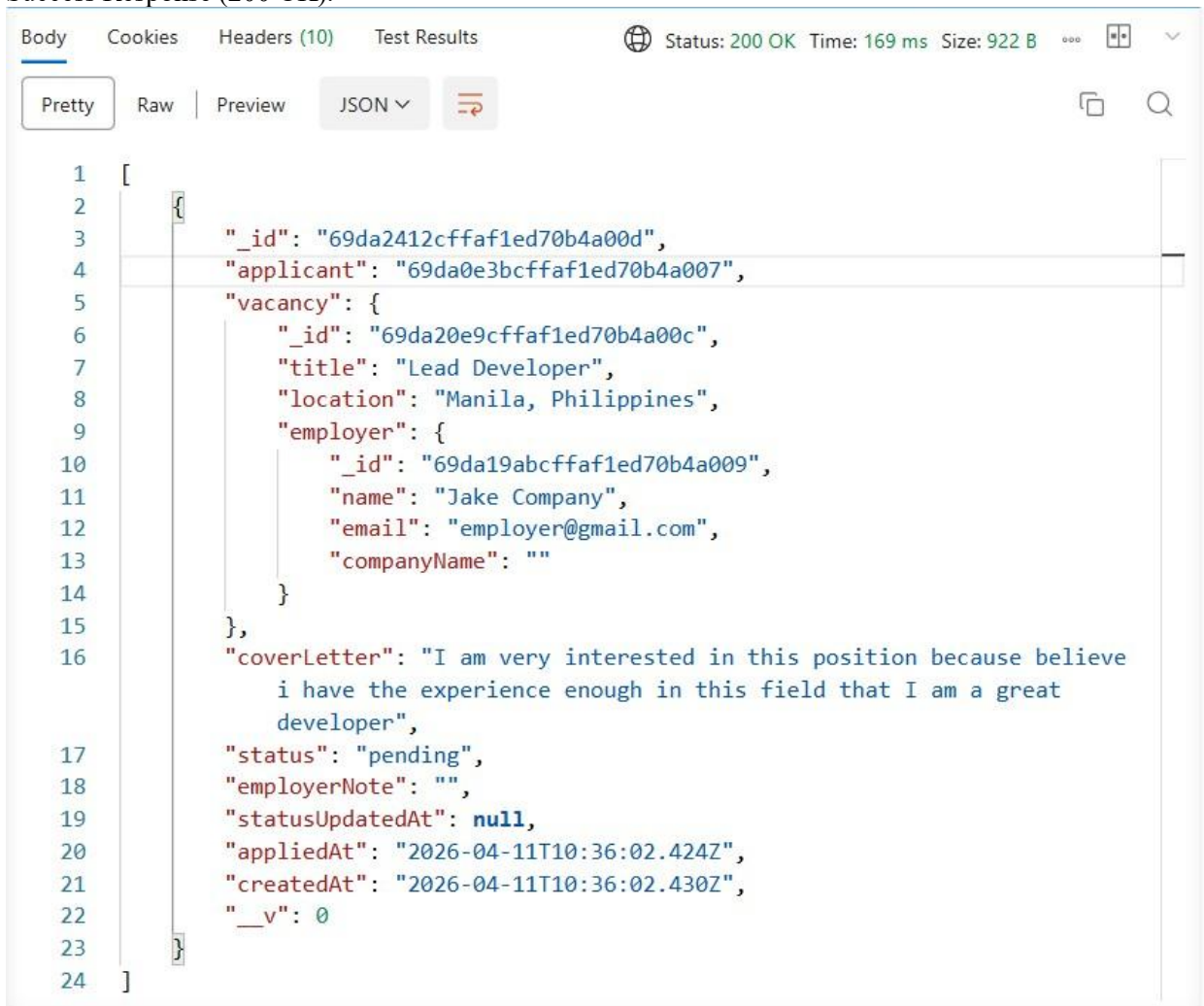
Params Authorization Headers (8) Body Scripts Settings Code Cookies

Type: Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#)

Token: eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5ZGEwZTNiY2ZmYWYxZWQ3MGI0YTAwNyIsInVjbGUiOiJyZXNpZGVudCIsImhhdCI6MTc3NTkwNDY5NywiZXhwIjoxNzc4NDk2Njk3fQ.A8zroHBrKua5gb4QvjJGi7EfQUqMEZ1zVKYetA158bg

Success Response (200 OK):



Body Cookies Headers (10) Test Results Status: 200 OK Time: 169 ms Size: 922 B

Pretty Raw Preview JSON

```
[
  {
    "_id": "69da2412cfffaf1ed70b4a00d",
    "applicant": "69da0e3bcfffaf1ed70b4a007",
    "vacancy": {
      "_id": "69da20e9cfffaf1ed70b4a00c",
      "title": "Lead Developer",
      "location": "Manila, Philippines",
      "employer": {
        "_id": "69da19abcfffaf1ed70b4a009",
        "name": "Jake Company",
        "email": "employer@gmail.com",
        "companyName": ""
      }
    },
    "coverLetter": "I am very interested in this position because believe i have the experience enough in this field that I am a great developer",
    "status": "pending",
    "employerNote": "",
    "statusUpdatedAt": null,
    "appliedAt": "2026-04-11T10:36:02.424Z",
    "createdAt": "2026-04-11T10:36:02.430Z",
    "__v": 0
  }
]
```

3. Update Application

Method & URL: PUT /jobs/applications/:id

Authorization Required: Yes (Resident only)

Request Format:

The screenshot shows a REST client interface with the following details:

- Method:** PUT
- URL:** http://localhost:3000/api/jobs/applications/69da2412cffaf1ed70b4a00d
- Send Button:** A blue button labeled "Send".
- Tabs:** Params, Authorization, Headers (10), Body, Scripts, Settings. The "Body" tab is selected.
- Request Format:** none, form-data, x-www-form-urlencoded, raw (selected), binary, GraphQL, JSON.
- Body Content:**

```
1 {
2   "coverLetter": "I am very interested in this position, I needed it "
3 }
```
- Buttons:** Beautify

Success Response (200 OK):

The screenshot shows the response body of the PUT request in a REST client interface:

- Tabs:** Body, Cookies, Headers (10), Test Results. The "Body" tab is selected.
- Status:** 200 OK
- Time:** 318 ms
- Size:** 908 B
- Buttons:** Pretty, Raw, Preview, JSON, and a search icon.
- Body Content:**

```
1 {
2   "message": "Application updated successfully",
3   "application": {
4     "_id": "69da2412cffaf1ed70b4a00d",
5     "applicant": "69da0e3bcffaf1ed70b4a007",
6     "vacancy": {
7       "_id": "69da20e9cffaf1ed70b4a00c",
8       "title": "Lead Developer",
9       "location": "Manila, Philippines",
10      "employer": {
11        "_id": "69da19abcffaf1ed70b4a009",
12        "name": "Jake Company",
13        "email": "employer@gmail.com",
14        "companyName": ""
15      }
16    },
17    "coverLetter": "I am very interested in this position, I needed it ",
18    "status": "pending",
19    "employerNote": "",
20    "statusUpdatedAt": null,
21    "appliedAt": "2026-04-11T10:36:02.424Z",
22    "createdAt": "2026-04-11T10:36:02.430Z",
23    "__v": 0
24  }
25 }
```

4. Delete Application

Method & URL: DELETE /jobs/applications/:id

Authorization Required: Yes (Resident only)

The screenshot shows a REST client interface with the following details:

- Method:** DELETE
- URL:** http://localhost:3000/api/jobs/applications/69da2412cffaf1ed70b4a00d
- Authorization:** Bearer Token (selected)
- Token:** eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5ZGEwZTNiY2ZmYWYxZWQ3MGI0YTAwNyIsInJvbGUiOiJyZXNpZGVudCIsImhhdCI6MTc3NTkwNDY5NywiZXhwIjoxNzc4NDk2Njk3fQ.A8zroH8rKua5gb4QvjGi7EfQUqMEZ1zVKYeta158bg

Success Response (200 OK):

The screenshot shows the response body of the DELETE request:

```
{
  "message": "Application withdrawn successfully",
  "id": "69da2412cffaf1ed70b4a00d"
}
```

Response details: Status: 200 OK, Time: 355 ms, Size: 421 B.

5. Get Applications for Job

Method & URL: GET /jobs/:id/applications

Authorization Required: Yes (Employer - Owner or Admin)

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** http://localhost:3000/api/jobs/69da20e9cffaf1ed70b4a00c/applications
- Authorization:** Bearer Token (selected)
- Token:** eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVC...

Success Response (200 OK):

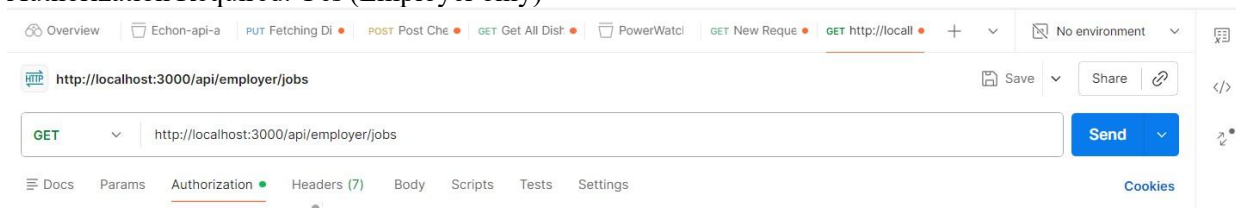


EMPLOYER ENDPOINTS

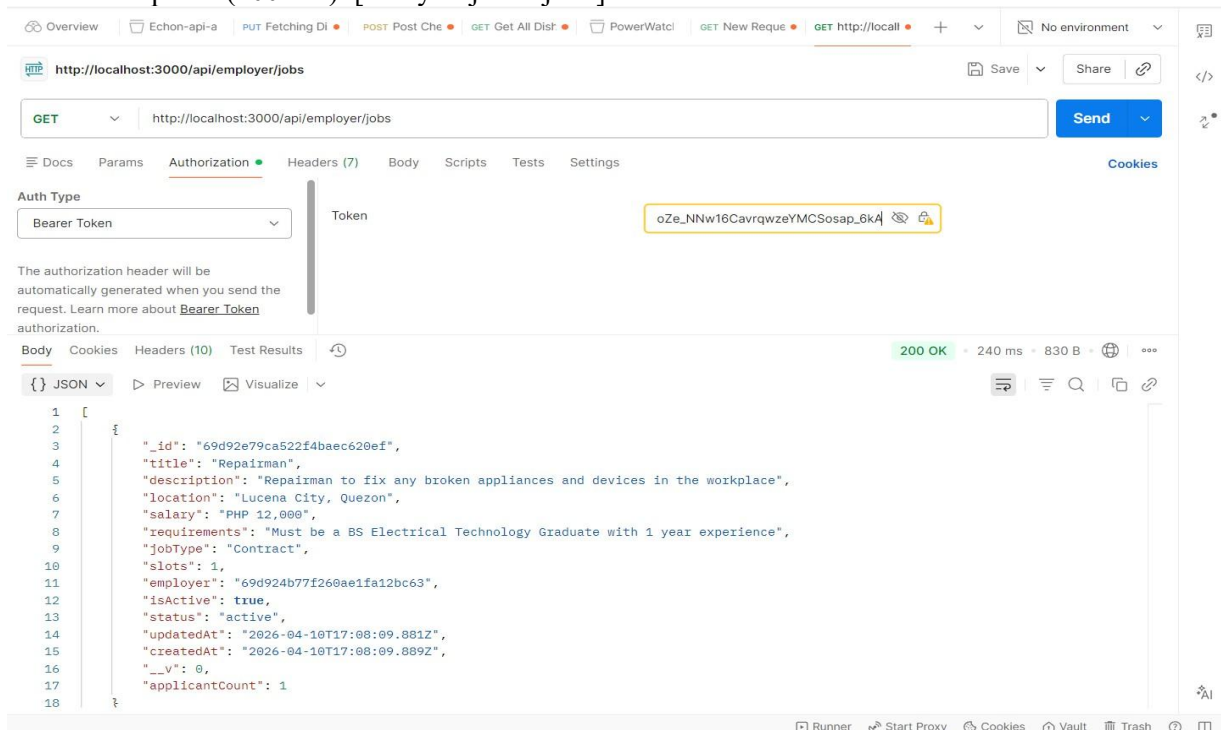
1. Get Employer's Jobs

Method & URL: GET /employer/jobs

Authorization Required: Yes (Employer only)



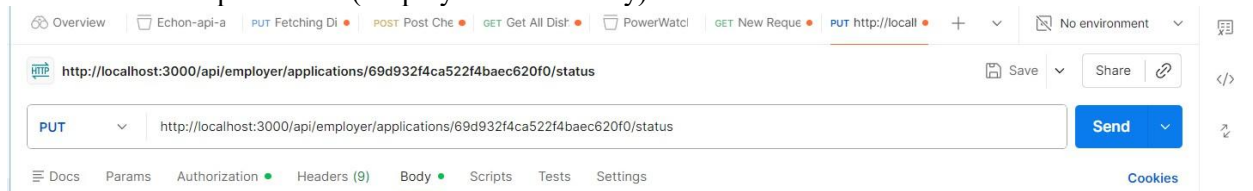
Success Response (200 OK): [Array of job objects]



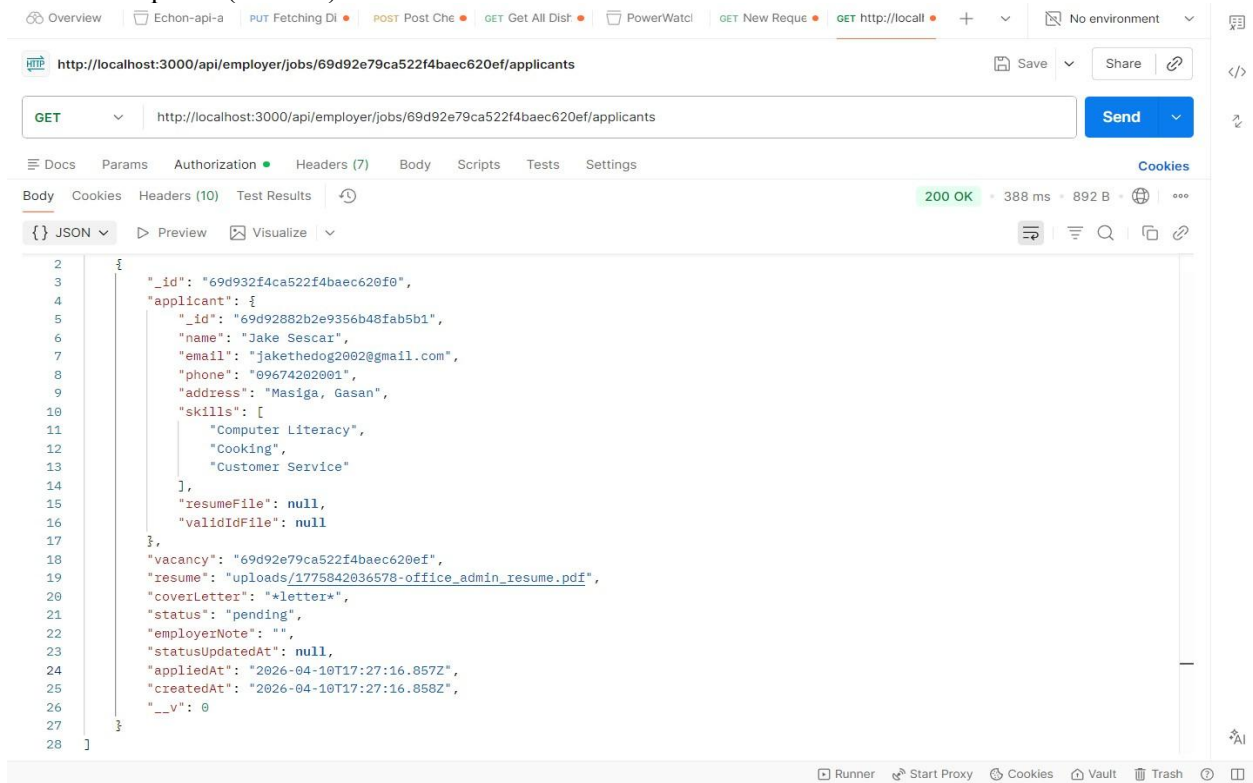
2. Get Applicants for Job

Method & URL: GET /employer/jobs/:jobId/applicants

Authorization Required: Yes (Employer - Owner only)



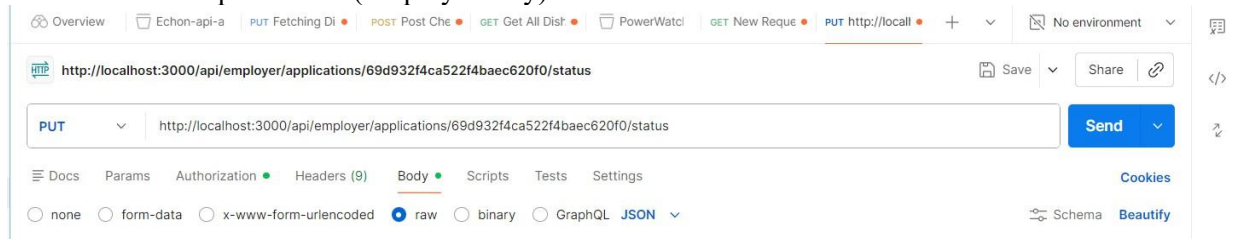
Success Response (200 OK):



3. Update Application Status

Method & URL: PUT /employer/applications/:applicationId/status

Authorization Required: Yes (Employer only)



Request Format:

Allowed statuses: pending, reviewed, shortlisted, rejected, hired



Success Response (200 OK):

The screenshot shows a REST client interface with the following details:

- URL:** `http://localhost:3000/api/employer/applications/69d932f4ca522f4baec620f0/status`
- Method:** PUT
- Body:**

```
{
  "status": "shortlisted",
  "employerNote": "Please review the candidate's portfolio."
}
```
- Response:** 200 OK, 1.20 s, 1017 B
- Response Body (JSON):**

```
{
  "_id": "69d932f4ca522f4baec620f0",
  "applicant": {
    "_id": "69d92882b2e9356b48fab5b1",
    "name": "Jake Sescar",
    "email": "jakethedog2002@gmail.com",
    "phone": "09674202001",
    "address": "Masiga, Gasan",
    "skills": [
      "Computer Literacy",
      "Cooking",
      "Customer Service"
    ],
    "resumeFile": null,
    "validIdFile": null
  },
  "vacancy": {
    "_id": "69d92e79ca522f4baec620ef",
    "title": "Repairman",
  }
}
```

4. Get Employer Statistics

Method & URL: GET /employer/stats

The screenshot shows a REST client interface with the following details:

- URL:** `http://localhost:3000/api/employer/stats`
- Method:** GET
- Body:** (Empty)
- Response:** (Not yet received)

Authorization Required: Yes (Employer only)
Success Response (200 OK):

The screenshot shows a REST client interface with the following details:

- URL:** `http://localhost:3000/api/employer/stats`
- Method:** GET
- Body:**

```
{
  "status": "shortlisted",
  "employerNote": "Please review the candidate's portfolio."
}
```
- Response:** 200 OK, 587 ms, 435 B
- Response Body (JSON):**

```
{
  "totalJobs": 1,
  "activeJobs": 1,
  "totalApplicants": 1,
  "pendingReview": 0,
  "shortlisted": 1,
  "hired": 0
}
```

5. Get Employer Profile Stats

Method & URL: GET /employer/profile-stats

The screenshot shows a REST client interface with the following details:

- URL:** `http://localhost:3000/api/employer/profile-stats`
- Method:** GET
- Body:**

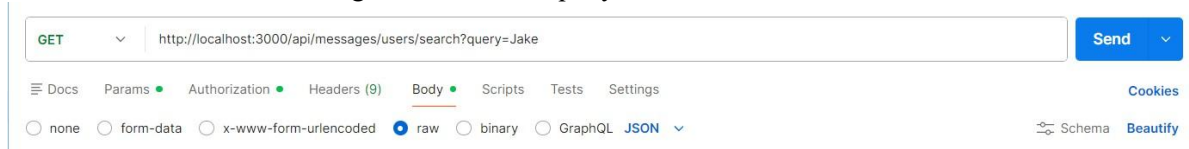
```
{
  "status": "shortlisted",
  "employerNote": "Please review the candidate's portfolio."
}
```
- Response:** 200 OK, 614 ms, 392 B
- Response Body (JSON):**

```
{
  "activeJobs": 1,
  "totalApplicants": 1,
  "closedJobs": 0
}
```

MESSAGING ENDPOINTS

1. Search Users

Method & URL: GET /messages/users/search?query=Jake



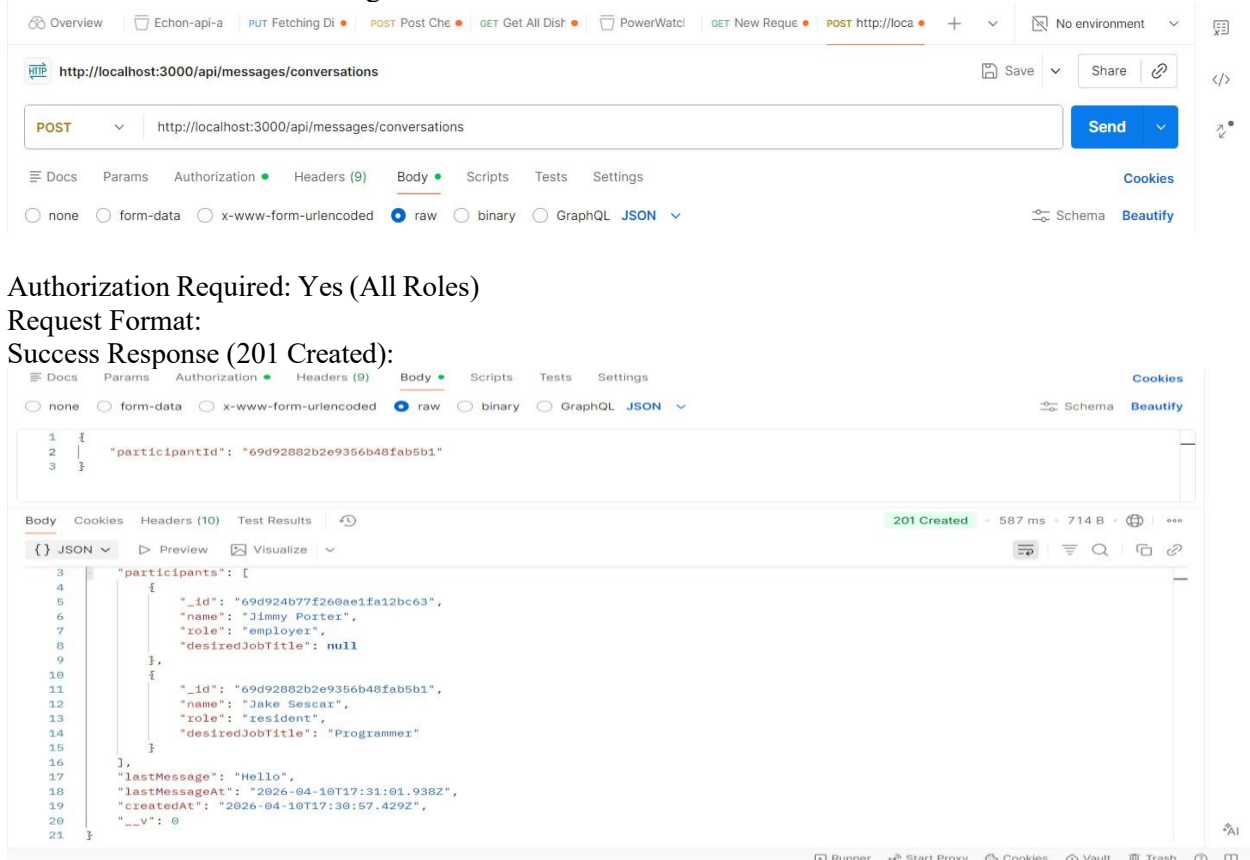
Authorization Required: Yes (All Roles)

Success Response (200 OK):



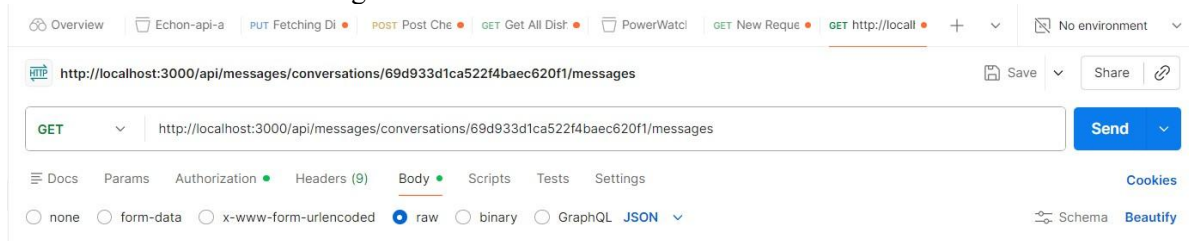
2. Create Conversation

Method & URL: POST /messages/conversations



3. Get Conversations

Method & URL: GET /messages/conversations



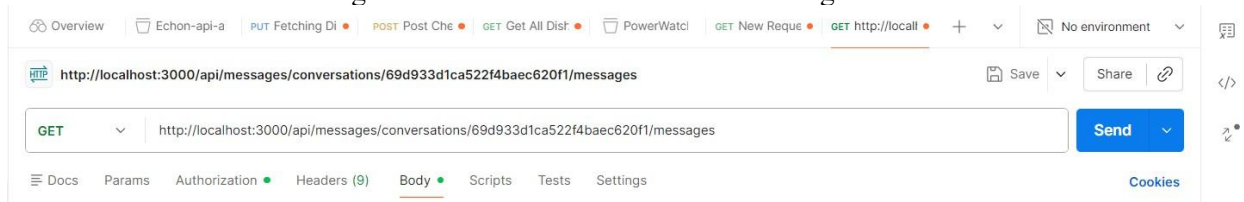
Authorization Required: Yes (All Roles)

Success Response (200 OK):



4. Get Messages in Conversation

Method & URL: GET /messages/conversations/:conversationId/messages



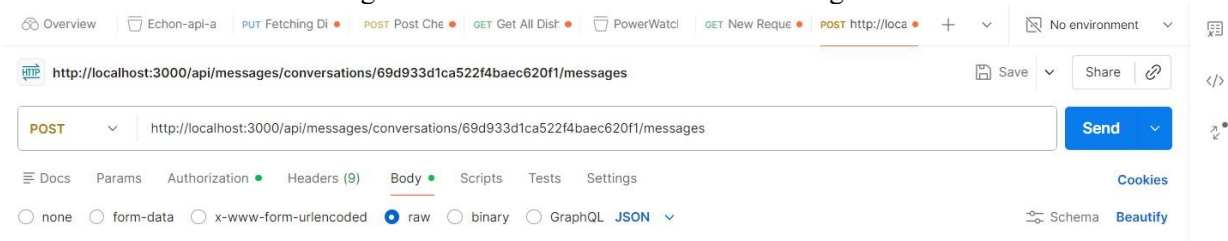
Authorization Required: Yes (Participant)

Success Response (200 OK):



5. Send Message

Method & URL: POST /messages/conversations/:conversationId/messages



Authorization Required: Yes (Participant)

Request Format:

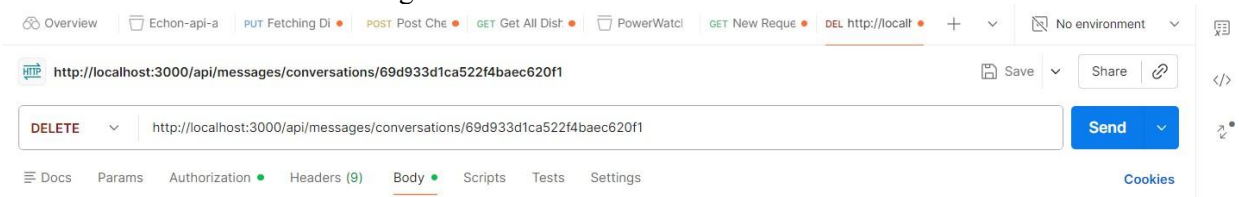


Success Response (200 OK):



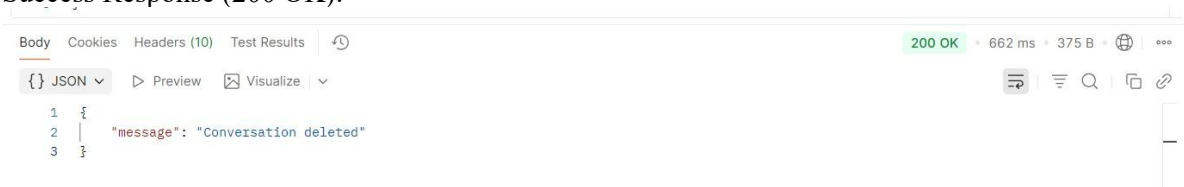
6. Delete Conversation

Method & URL: DELETE /messages/conversations/:conversationId



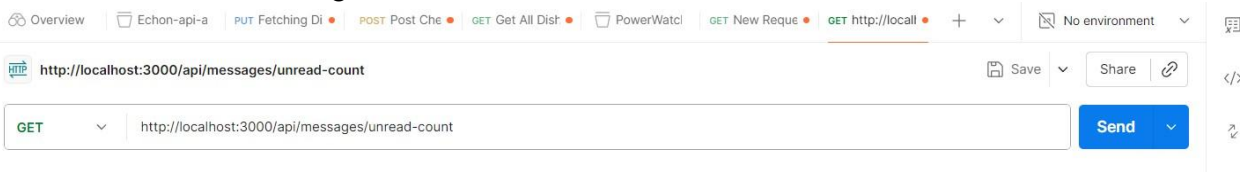
Authorization Required: Yes (Participant)

Success Response (200 OK):



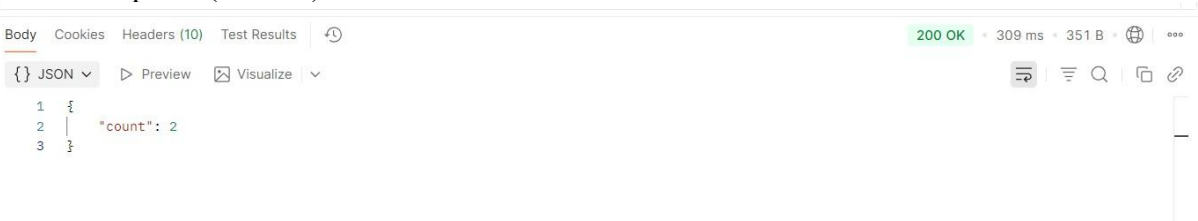
7. Get Unread Message Count

Method & URL: GET /messages/unread-count



Authorization Required: Yes (All Roles)

Success Response (200 OK):

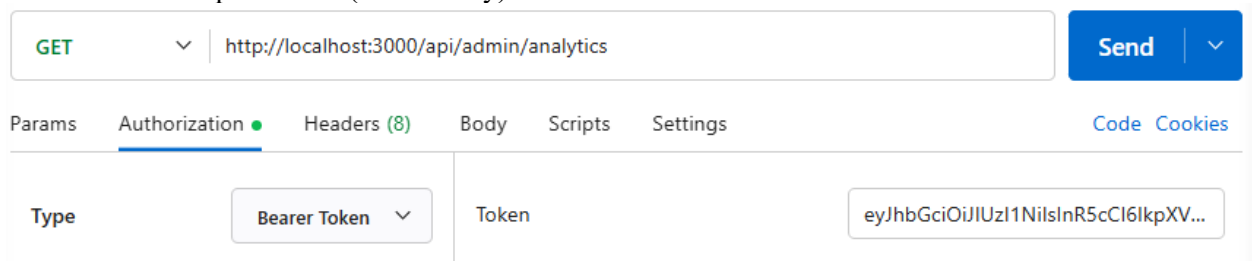


ADMIN ENDPOINTS

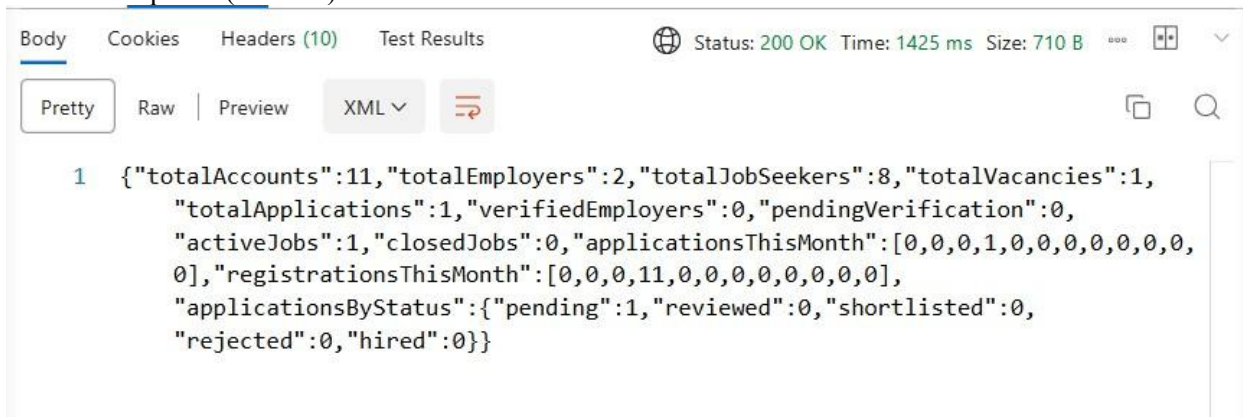
1. Get Analytics

Method & URL: GET /admin/analytics

Authorization Required: Yes (Admin only)



Success Response (200 OK):



```
Body Cookies Headers (10) Test Results Status: 200 OK Time: 1425 ms Size: 710 B
Pretty Raw Preview JSON
1 {
2   "totalAccounts": 11,
3   "totalEmployers": 2,
4   "totalJobSeekers": 8,
5   "totalVacancies": 1,
6   "totalApplications": 1,
7   "verifiedEmployers": 0,
8   "pendingVerification": 0,
9   "activeJobs": 1,
10  "closedJobs": 0,
11  "applicationsThisMonth": [
12    0,
13    0,
14    0,
15    1,
16    0,
17    0,
18    0,
19    0,
20    0,
21    0,
22    0,
23    0
24  ],
25  "registrationsThisMonth": [
26    0,
27    0,
28    0,
29    11,
30    0,
31    0,
32    0,
33    0,
```

2. Get All Users

Method & URL: GET /admin/users?limit=50&page=1

Authorization Required: Yes (Admin only)

Query Parameters: ?role=employer&verificationStatus=pending&limit=50&page=1

GET http://localhost:3000/api/admin/users?limit=50&page=1 Send

Params Authorization Headers (8) Body Scripts Settings Code Cookies

Type Bearer Token

Token eyJhbGciOiJIUzI1NiIsInR5cCI6IkpX...

Success Response (200 OK):

Body Cookies Headers (10) Test Results  Status: 200 OK Time: 332 ms Size: 2.8 KB ...  

Pretty Raw Preview JSON   

```
1 {
2   "users": [
3     {
4       "_id": "69da19abcfaf1ed70b4a009",
5       "name": "Jake Company",
6       "email": "employer@gmail.com",
7       "role": "employer",
8       "verificationStatus": "unverified",
9       "isActive": true,
10      "hasCompletedOnboarding": false,
11      "createdAt": "2026-04-11T09:51:39.268Z"
12    },
13    {
14      "_id": "69da0e3bcffaf1ed70b4a007",
15      "name": "Alkimar Guitang",
16      "email": "AlkimarG@gmail.com",
17      "role": "resident",
18      "verificationStatus": "unverified",
19      "isActive": true,
20      "hasCompletedOnboarding": false,
21      "createdAt": "2026-04-11T09:02:51.846Z"
22    },
23    {
24      "isActive": true,
25      "hasCompletedOnboarding": false,
26      "_id": "69d93114b15909826c83f220",
27      "name": "lovely",
28      "email": "lovely@gmail.com",
29      "role": "employer",
30      "verificationStatus": "unverified",
31      "createdAt": "2026-04-10T17:19:16.733Z"
32    }
33  ]
34 }
```

3. Update User Role

Method & URL: PUT /admin/users/:id/role

Authorization Required: Yes (Admin only)

Request Format:

The screenshot shows a REST client interface with the following details:

- Method:** PUT
- URL:** http://localhost:3000/api/admin/users/69d93114b15909826c83f220/role
- Body:** A JSON object with the following fields:

```
{ 1: { 2:   "role": "employer", 3:   "companyName": "Hard Copy", 4:   "industry": "Production" 5: }
```
- Format:** JSON
- Buttons:** Send, Beautify

Success Response (200 OK):

The screenshot shows the response body of the PUT request, which is a JSON object:

```
1 { "message": "User role updated", "user": { "isActive": true, 2:   "hasCompletedOnboarding": false, "_id": "69d93114b15909826c83f220", "name": "lovely", 3:   "email": "lovely@gmail.com", "password": "$2b$10$D31Yb2sR5Hq/ 4:   texOtqItUebFfeMtY9A7bCWc.A.jXmTnRYCA003Xe", "role": "employer", "about": "", 5:   "phone": null, "dateOfBirth": null, "gender": null, "address": null, "skills": [], 6:   "desiredJobTitle": null, "educationalAttainment": null, "workExperience": null, 7:   "availabilityStatus": null, "companyName": "", "industry": "", "companySize": "", 8:   "website": "", "companyDescription": "", "businessAddress": "", 9:   "verificationStatus": "unverified", "businessPermitUrl": null, 10:  "registrationDocUrl": null, "resumeFile": null, "validIdFile": null, 11:  "onboardingComplete": false, "createdAt": "2026-04-10T17:19:16.733Z", "__v": 0 } }
```

4. Deactivate User

Method & URL: PUT /admin/users/:id/deactivate

Authorization Required: Yes (Admin only)

Request Format:

The screenshot shows a REST client interface with the following details:

- Method:** PUT
- URL:** http://localhost:3000/api/admin/users/69d92d24b15909826c83f21a/deactivate
- Body:** A JSON object with the following field:

```
{ 1: { 2:   "reason": "Violation of terms of service" 3: }
```
- Format:** JSON
- Buttons:** Send, Beautify

Success Response (200 OK):

Body Cookies Headers (10) Test Results Status: 200 OK Time: 432 ms Size: 1.07 KB

Pretty Raw Preview JSON

```
1 [{"message": "User deactivated", "user": {"isActive": false, "hasCompletedOnboarding": false, "_id": "69d92d24b15909826c83f21a", "name": "Dimagiba Tulog", "email": "dimagiba@gmail.com", "password": "$2b$10$u6a5S4pXJNehFJOAvTZ1T.iEpEiKHZYAAaVZ2FSrjaLF5wwsPWa60", "role": "resident", "about": "", "phone": null, "dateOfBirth": null, "gender": null, "address": null, "skills": [], "desiredJobTitle": null, "educationalAttainment": null, "workExperience": null, "availabilityStatus": null, "companyName": "", "industry": "", "companySize": "", "website": "", "companyDescription": "", "businessAddress": "", "verificationStatus": "unverified", "businessPermitUrl": null, "registrationDocUrl": null, "resumeFile": null, "validIdFile": null, "onboardingComplete": false, "createdAt": "2026-04-10T17:02:28.064Z", "__v": 0}}]
```

5. Reactivate User

Method & URL: PUT /admin/users/:id/reactivate

Authorization Required: Yes (Admin only)

PUT http://localhost:3000/api/admin/users/69d92d24b15909826c83f21a/reactivate Send

Params Authorization Headers (10) Body Scripts Settings Code Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Beautify

```
1 [{"message": "User reactivated successfully", "user": {"_id": "69d92d24b15909826c83f21a", "isActive": true}}]
```

Success Response (200 OK):

Body Cookies Headers (10) Test Results Status: 200 OK Time: 387 ms Size: 1.07 KB

Pretty Raw Preview JSON

```
1 {
2   "message": "User reactivated",
3   "user": {
4     "_id": "69d92d24b15909826c83f21a",
5     "name": "Dimagiba Tulog",
6     "email": "dimagiba@gmail.com",
7     "password": "$2b$10$u6a5S4pXJNehFJOAvTZ1T.iEpEiKHZYAAaVZ2FSrjaLF5wvsPwa60",
8     "role": "resident",
9     "about": "",
10    "phone": null,
11    "dateOfBirth": null,
12    "gender": null,
13    "address": null,
14    "skills": [],
15    "desiredJobTitle": null,
16    "educationalAttainment": null,
17    "workExperience": null,
18    "availabilityStatus": null,
19    "companyName": "",
20    "industry": "",
21    "companySize": "",
22    "website": "",
23    "companyDescription": "",
24    "businessAddress": "",
25    "verificationStatus": "unverified",
26    "businessPermitUrl": null,
27    "registrationDocUrl": null,
28    "resumeFile": null,
29    "validIdFile": null,
30    "onboardingComplete": false
```

6. Update Verification Status

Method & URL: PUT /admin/users/:id/verification

Authorization Required: Yes (Admin only)

Request Format:

PUT http://localhost:3000/api/admin/users/69d93114b15909826c83f220/verification Send

Params Authorization Headers (10) Body Scripts Settings Code Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Beautify

```
1 {
2   "verificationStatus": "verified"
3 }
```

Success Response (200 OK):

Body Cookies Headers (10) Test Results Status: 200 OK Time: 216 ms Size: 1.07 KB

Pretty Raw Preview JSON

```
1 {
2   "message": "Employer verification updated",
3   "user": {
4     "_id": "69d93114b15909826c83f220",
5     "name": "lovely",
6     "email": "lovely@gmail.com",
7     "password": "$2b$10$D31Yb2sR5Hq/tex0tqItUebFfeMtY9A7bCWc.A.jXmTNRyCA003Xe",
8     "role": "employer",
9     "about": "",
10    "phone": null,
11    "dateOfBirth": null,
12    "gender": null,
13    "address": null,
14    "skills": [],
15    "desiredJobTitle": null,
16    "educationalAttainment": null,
17    "workExperience": null,
18    "availabilityStatus": null,
19    "companyName": "",
20    "industry": "",
21    "companySize": "",
22    "website": "",
23    "companyDescription": "",
24    "businessAddress": "",
25    "verificationStatus": "verified",
26    "businessPermitUrl": null,
```

7. Delete User

Method & URL: DELETE /admin/users/:id

Authorization Required: Yes (Admin only)

DELETE http://localhost:3000/api/admin/users/69d92ba1b15909826c83f217 Send

Params Authorization Headers (8) Body Scripts Settings Code Cool

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL Text

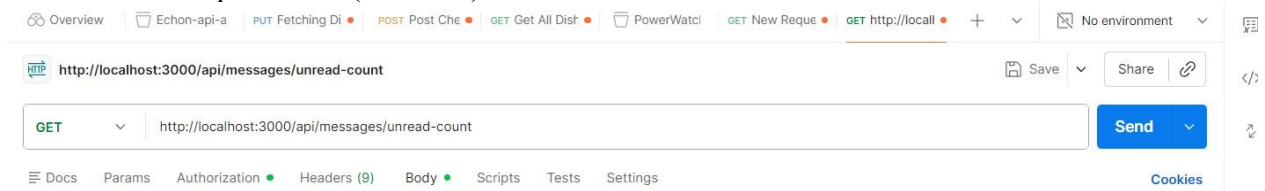
```
1 {
2   "message": "User deleted successfully"
3 }
```

USER ENDPOINTS

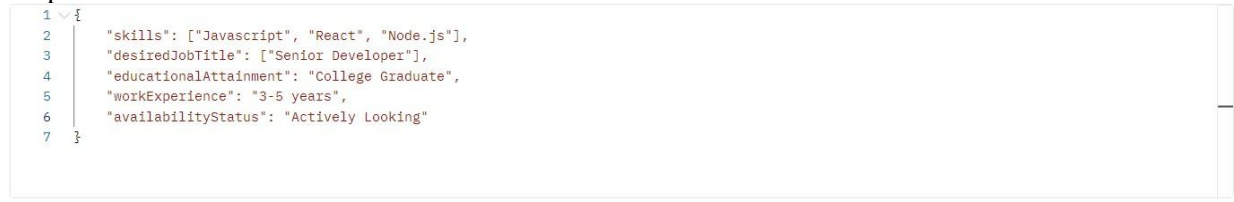
1. Complete Onboarding

Method & URL: PUT /users/onboarding

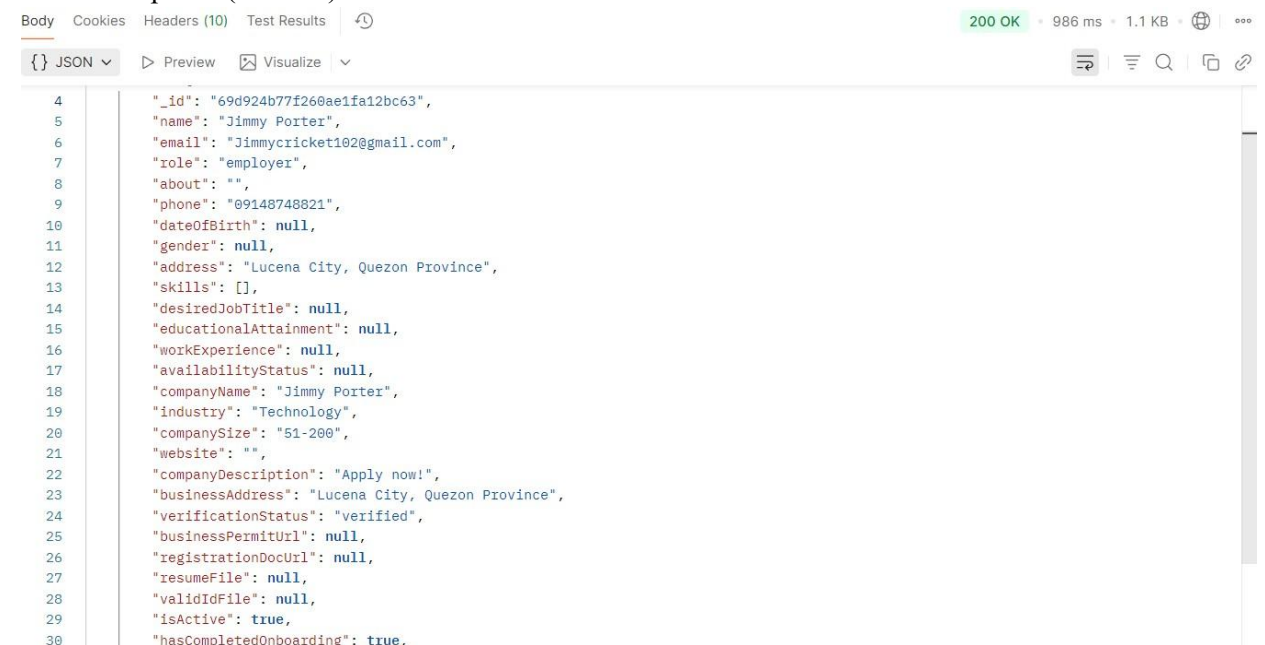
Authorization Required: Yes (All Roles)



Request Format:



Success Response (200 OK):



3.4 Advanced API Features

Pagination, Filtering & Sorting

Job Search Example:

```
GET /api/v1/jobs?limit=20&page=2&location=Marinduque&jobType=Full-time&sort=-createdAt
```

Query Parameters:

- limit: Number of records per page (default: 20, max: 100)
- page: Page number (default: 1)
- sort: Sort field (prefix with - for descending, + for ascending)
- search: Search term for job titles and descriptions

- location: Filter by job location (e.g., Manila, Remote)
- jobType: Filter by employment type (Full-time, Part-time, Contract, Freelance)
- category: Filter by job category (e.g., IT, Sales, Marketing)
- minSalary: Minimum salary filter
- maxSalary: Maximum salary filter

User Search Example:

```
GET/api/v1/admin/users?role=employer&verificationStatus=pending&isActive=true&limit=50&page=1&sort=-createdAt
```

Response Format with Pagination

```
{
  "message": "Jobs retrieved successfully",
  "data": {
    "items": [
      {
        "_id": "507f1f77bcf86cd799439011",
        "jobTitle": "Senior Developer",
        "salary": 80000,
        "location": "Marinduque",
        "createdAt": "2025-01-15T10:30:00Z"
      }
    ],
    "pagination": {
      "currentPage": 2,
      "totalPages": 5,
      "totalRecords": 100,
      "recordsPerPage": 20,
      "hasNextPage": true,
      "hasPreviousPage": true
    }
  }
}
```

File Upload Handling

- Allowed types: PDF, DOC, DOCX, JPEG, JPG, PNG, GIF
- Max size: 10MB per file
- Content-Type: multipart/form-data

Versioning Strategy

- Current: /api/v1
- Backward Compatibility: /api (aliases to v1)
- Deprecation: 6-month notice before removal

3.5 Testing Tools & Examples

Recommended Testing Tools

1. Postman - GUI-Based API Testing
 - o Download: <https://www.postman.com/downloads/>
 - o Features: Collections, Environments, Pre-request scripts
2. cURL - Command-Line HTTP Client
 - o Built-in on Mac/Linux, or download from <https://curl.se/>
3. Newman - Postman CLI Automation

```
npm install -g newman
newman run docs/postman-collection.json --environment docs/env.json
```

4. Insomnia - Lightweight REST Client
 - o Download: <https://insomnia.rest/>

Sample cURL Examples

1. Register a New User

```
//POST http://localhost:5000/api/v1/auth/register \
//"Content-Type: application/json" \
'{
  "name": "Alkimar Guitang",
  "email": "AlkimarGuitang@example.com",
  "password": "MyPassword123"
}'
```

2. Login User

```
// POST http://localhost:5000/api/v1/auth/login \
//"Content-Type: application/json" \
{
  "email": "AlkimarGuitang@example.com",
  "password": "MyPassword123"
}'
```

CHAPTER 4: DEPLOYMENT & LINKS

4.1 Live Application Links.

- Frontend Web Application URL: <https://strampeso.vercel.app/>
- Backend API Base URL: <https://dashboard.render.com/web/srv-d7dl35l8nd3s73b49stg>
- GitHub Master Repository URL: <https://github.com/echonalexis-web/Stram-Peso>

4.2 Testing Coverage Summary

- Unit and Integration Tests

All files	0	0	0	0	
controllers	0	0	0	0	
adminController.js	0	0	0	0	1-270
authController.js	0	0	0	0	1-357
employerController.js	0	0	0	0	1-274
jobController.js	0	0	0	0	1-291
messageController.js	0	0	0	0	1-266
userController.js	0	0	0	0	1-51
middleware	0	0	0	0	
auth.js	0	0	0	0	1-75
models	0	100	100	0	
Conversation.js	0	100	100	0	1-12
InviteCode.js	0	100	100	0	1-12
JobApplication.js	0	100	100	0	1-29
JobVacancy.js	0	100	100	0	1-22
Message.js	0	100	100	0	1-13
User.js	0	100	100	0	1-74
routes	0	0	0	0	
adminRoutes.js	0	100	100	0	1-21
messageController.js	0	0	0	0	1-266
userController.js	0	0	0	0	1-51
middleware	0	0	0	0	
auth.js	0	0	0	0	1-75
models	0	100	100	0	
Conversation.js	0	100	100	0	1-12
InviteCode.js	0	100	100	0	1-12
JobApplication.js	0	100	100	0	1-29
JobVacancy.js	0	100	100	0	1-22
Message.js	0	100	100	0	1-13
User.js	0	100	100	0	1-74
routes	0	0	0	0	
adminRoutes.js	0	100	100	0	1-21
adminRoutes.js	0	100	100	0	1-21
authRoutes.js	0	100	0	0	1-67
employerRoutes.js	0	100	100	0	1-23
jobRoutes.js	0	0	0	0	1-55
messageRoutes.js	0	100	100	0	1-21
userRoutes.js	0	100	100	0	1-7

Test Suites: 1 passed, 1 total					
Tests: 1 passed, 1 total					
Snapshots: 0 total					
Time: 4.045 s					
Ran all test suites.					

- End-to-End (E2E) Tests:

Iteration 1

POST 1) Admin Login

http://localhost:3000/api/v1/auth/login

200 • 357 ms • 664 B • 2

PASS Admin login success

PASS Admin token exists

POST 2) Generate Invite Code (Admin)

http://localhost:3000/api/v1/auth/invite

200 • 177 ms • 308 B • 2

PASS Invite code generated

PASS Invite code exists

POST 3) Register Employer

http://localhost:3000/api/v1/auth/register/employer

200 • 508 ms • 335 B • 1

PASS Employer register success

POST 4) Register Resident

http://localhost:3000/api/v1/auth/register

200 • 354 ms • 764 B • 1

PASS Resident register success

POST 5) Employer Login

http://localhost:3000/api/v1/auth/login

200 • 249 ms • 679 B • 1

PASS Employer login success

POST 6) Resident Login

http://localhost:3000/api/v1/auth/login

200 • 212 ms • 665 B • 1

PASS Resident login success

POST 7) Employer Create Job

http://localhost:3000/api/v1/employer/jobs

201 • 200 ms • 662 B • 2

PASS Create job success

PASS Job id captured

POST 8) Resident Apply to Job

http://localhost:3000/api/v1/jobs/69dc9592c8250b61555a477d/apply

200 • 586 ms • 682 B • 1

PASS Apply success

GET 9) Employer Get Applicants For Job

http://localhost:3000/api/v1/employer/jobs/69dc9592c8250b61555a477d/applicants

200 • 526 ms • 756 B • 2

PASS Applicants fetched

PASS Applicants is array

PUT 10) Employer Update Application Status

http://localhost:3000/api/v1/employer/applications/69dc9593c8250b61555a477e/status

200 • 1943 ms • 876 B • 1

PASS Status updated

POST 11) Employer Create Conversation With Resident

http://localhost:3000/api/v1/messages/conversations

201 • 540 ms • 757 B • 2

PASS Conversation created

PASS Conversation id captured

POST 12) Employer Send Message

http://localhost:3000/api/v1/messages/conversations/69dc9595c8250b61555a477f/messages

201 • 600 ms • 574 B • 1

PASS Message sent

GET 13) Resident Get Messages

http://localhost:3000/api/v1/messages/conversations/69dc9595c8250b61555a477f/messages

200 • 450 ms • 797 B • 2

PASS Messages fetched

PASS Messages array

The STRAM PESO system was validated through a 13-step end-to-end Postman workflow covering all major user roles (Admin, Employer, Resident) and all core features (auth, jobs, applications, messaging). All requests returned expected status codes and payloads. Unit test infrastructure is in place with model and route layers confirmed at 100% function and branch coverage.

REFERENCES

- [1] United Nations. Goal 8: Economic Growth – United Nations Sustainable Development. Available at: <https://sdgs.un.org/goals/goal8> (accessed March 18, 2026).
- [2] World Bank. 2025. Rethinking Public Employment Services for the Digital Era. World Bank Blogs. Available at: <https://blogs.worldbank.org/en/jobs/rethinking-public-employment-services-digital-era> (accessed March 18, 2026).
- [3] International Labour Organization (ILO). Global Employment Policy Review 2025. ILO Publications, 2025. Available at: <https://www.ilo.org/global/publication> (ilo.org in Bing) (accessed March 18, 2026).
- [4] International Labour Organization (ILO). Global Employment Policy Review 2025. ILO Publications, 2025. Available at: <https://www.ilo.org/global/publications> (ilo.org in Bing) (accessed March 18, 2026).
- [5] Graham, M., Ferrari, F. (eds.). Digital Work in the Planetary Market. MIT Press, 2022. DOI: 10.7551/mitpress/13835.001.0001.
- [6] Bureau of Labor Statistics (U.S.). The Rise in Remote Work Since the Pandemic and Its Impact on Employment. Washington, DC: BLS, 2023.
- [7] Provincial Government of Marinduque. Employment – Livelihood Manpower Development Public Employment Service Office (LMD-PESO). Official Website of the Province of Marinduque. Available at: <https://marinduque.gov.ph> (accessed March 18, 2026).
- [8] SEEK Asia. 2026. About JobStreet (Philippines). Available at: <https://ph.jobstreet.com/about/en>
- [9] Indeed Editorial Team. 2026. How Does Indeed Work? Available at: <https://www.indeed.com/career-advice/finding-a-job/how-does-indeed-work>
- [10] LinkedIn Corporation. 2025. LinkedIn Recruiter Features. Available at: <https://business.linkedin.com/talent-solutions/recruiter/recruiter-features>
- [11] Philippines-Jobs.com. 2024. About Us. Available at: <https://www.philippines-jobs.com/home/about-us/>
- [12] Province of Marinduque. 2024. Employment Generation and Assistance (PESO Programs). <https://marinduque.gov.ph/job/>
- [13] Department of Labor and Employment (DOLE). 2024. PhilJobNet Official Portal. <https://philjobnet.gov.ph/>

- [14] Kalibrr. 2024. About Us – Kalibrr Employer Branding. <https://employerbranding.kalibrr.com/about-us/>
- [15] Foundit. 2022. Job platform Monster transforms to an end-to-end talent platform. Available at: <https://economictimes.indiatimes.com/tech/technology/jobsearch-portal-monster-rebranded-as-foundit-becomes-talent-management-portal/articleshow/95715997.cms>
- [16] Manila Recruitment. 2024. Corporate Headhunters in the Philippines (Headhunting Services Overview). Available at: <https://manilarecruitment.com/headhunters-in-the-philippines/>
- [17] Technical Education and Skills Development Authority (TESDA). 2024. About TESDA – Mission, Vision, and Programs. Available at: <https://www.tesda.gov.ph/About/TESDA/11>
- [18] Careerjet. 2024. About Us – Careerjet Job Search Engine. Available at: <https://www.careerjet.com/about-us>
- [19] OnlineJobs.ph. 2024. About Us – Culture, Values and Team. Available at: <https://v2.onlinejobs.ph/aboutus>
- [20] Bossjob. 2026. About Bossjob – AI Job Portal Philippines. Available at: <https://bossjob.ph/>
- [21] Glassdoor. 2024. About Glassdoor – Making Worklife Better Together. Available at: <https://www.glassdoor.com/about/>

APPENDICES

Appendix A: Glossary of API Terms

- API (Application Programming Interface)** – Refers to the set of rules and protocols used in the STRAM-PESO system that allows the frontend and backend to communicate and exchange data.
- Authentication** – Refers to the process in STRAM-PESO where users verify their identity through login credentials before accessing protected API endpoints.
- Authorization** – Refers to the process of determining whether a user has permission to access specific API endpoints based on their role.
- Controller** – Refers to the part of the backend that contains the logic for handling API requests and sending responses.
- CRUD Operations** – Refers to the basic database actions supported by the STRAM-PESO API: Create, Read, Update, and Delete.
- Database Query** – Refers to a request made to the MongoDB database to retrieve, insert, update, or delete data.
- Endpoint** – Refers to a specific URL in the STRAM-PESO API where requests are sent, such as `/api/auth/login` or `/api/jobs`.
- Environment Variables** – Refers to secure configuration values (e.g., database URI, JWT secret) stored in `.env` files and used by the system.

Error Handling – Refers to the mechanism used in STRAM-PESO to manage and respond to errors during API requests.

HTTP Methods – Refers to the types of actions used in API requests in STRAM-PESO, including GET (retrieve data), POST (create data), PUT (update data), and DELETE (remove data).

JSON (JavaScript Object Notation) – Refers to the data format used in STRAM-PESO for sending and receiving structured data between client and server.

Middleware – Refers to the functions in the STRAM-PESO backend that process requests before reaching the main logic, such as verifying tokens and handling errors.

Payload – Refers to the actual data included in a request or response, such as user information, job details, or application data.

Request – Refers to the message sent by the client (frontend) to the server (backend) to perform an action, such as retrieving or submitting data.

Response – Refers to the data returned by the server after processing a client request, usually in JSON format.

Route – Refers to the defined path in the backend that handles specific API requests and directs them to the appropriate controller.

Schema – Refers to the structure or model of data defined in MongoDB using Mongoose, such as User, Job, or Application schema.

Socket.IO – Refers to the technology used in STRAM-PESO to enable real-time communication features such as messaging.

Token – Refers to the secure string generated during login (JWT) that is used to verify and maintain a user's authenticated session.

Validation – Refers to the process of checking incoming request data to ensure it is complete, correct, and secure before processing.

Appendix B: Acronyms & Abbreviations

API – Application Programming Interface

ATS – Applicant Tracking System

CRUD – Create, Read, Update, Delete

CSS – Cascading Style Sheets

DB – Database

DOLE – Department of Labor and Employment

HTTP – Hypertext Transfer Protocol

JSON – JavaScript Object Notation

JWT – JSON Web Token

LGU – Local Government Unit

MERN – MongoDB, Express.js, React.js, Node.js

MVC – Model-View-Controller

NGO – Non-Governmental Organization

PESO – Public Employment Service Office

RBAC – Role-Based Access Control

REST – Representational State Transfer

SPA – Single Page Application

STRAM-PESO – Streamlined Recruitment and Applicant Management for Public Employment Service Office

UI – User Interface

UX – User Experience

UAT – User Acceptance Testing

Appendix C: Changelog / Document Revisions

DATE	AUTHOR(S)	CHANGE MADE
March 27 – April 3	All Members	Creation of PowerWatch as the project for Web System and Technologies 2
April 4, 2026	Alkimar Guitang	Changing PowerWatch project to STRAMPESO (although still retaining some features and platforms used by the recent project)
April 5, 2026	All members	Initial proposal draft submitted, chapter 1 completed
April 6, 2026	All Members	Added chapter 2
April 10, 2026	All Members	Added Chapter 3 and 4, references and revise the related literature 2
April 11, 2026	All Members	Updated testing strategy and MVP documentation
April 12, 2026	All Members	Finalized the appendices, -all deployments all links in vercel, MongoDB and GitHub

Appendix D: Proof of Validity / Certification

  	Republic of the Philippines MARINDUQUE STATE UNIVERSITY College of Information and Computing Sciences Panfilo M. Manguera, Sr. Rd., Tanza, Boac, Marinduque 4901 CICS Tel. No.: (042) 332 2853 CICS E-mail Address: cics@mscmarinduque.edu.ph Website: www.mscmarinduque.edu.ph
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March 12, 2026

MRS. ALMA TIMTIMAN
Chief Administrative Officer, LMD-PESO Marinduque
Capitol Compound, Santol, Boac, Marinduque

Dear Ma'am Timtiman,

Good Day!

As part of the requirements in our course, CAP301—Capstone Project 1 the degree Bachelor of Science in Information Technology (BSI/T), the undersigned, who are third-year students from the College of Information and Computing Sciences of Marinduque State University, would like to request your good office to allow us to conduct a preliminary investigation on the existing processes and operations in your organization, mainly in your Office. The course's primary requirement is to develop a project proposal that focuses on the course's underlying principles, drawing on the organization's identified problems.

As of now, we are conceptualizing the project entitled "STRAM-PESO: An Online Employment and Applicant Management System for Public Employment Service Office".

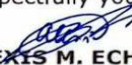
In this regard, may we earnestly ask your permission to please allow us to make your organization the subject of this undertaking and be able to do the following activities:

- Conduct data-gathering activities like interviews and observations about the day-to-day business transactions your organization delivers;
- Ask for background information about the organization and its business goals and objectives;
- Take pictures of the business setup or physical establishment;
- Gather sample forms used in daily transactions, especially related to this proposed study
- Conduct follow-up meetings through Face-to-Face, Google Meet, or Zoom for consultation, if necessary.

Rest assured that all data and information acquired while pursuing the study shall be treated with utmost confidentiality and only be used for the said purpose.

Your anticipated positive response regarding the matter will be highly appreciated.

Respectfully yours,

 ALEXIS M. ECHON  ALKIMAR S. GUITANG Proponents	 IVY L. CRUZADO  JAKE ROMAR I. SESCAR
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Noted-by:


ART JERVIN L. MAGCAMIT, LPT, MIT
Faculty-in-charge

LMD-PESO RECEIVED

DATE: 12 MAR 2026

TIME: 2:13 PM

BY: ETHEL J. ORILLA

CICS Goal: "To produce competitive IT professionals and IT-enabled individuals who will encourage real innovation for the advancement in the digital era of the province and the country as a whole."

MarSU Vision: "By 2030, MarSU is a globally recognized university, fostering academic excellence through research and innovation."

MarSU Mission: "MarSU provides excellence in instruction, research, extension and production, responsive to the needs of the 21st century education in pursuit of global transformation."

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